集成电路原理与设计

1.导论

谭志超
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Instructor: Zhichao Tan

- Education:
  - Ph.D.: Delft University of Technology 2013
  - M.S.: Peking University 2008
  - B. S.: Xi'an Jiaotong University 2004
- Working Experience :
  - Professor, Zhejiang University 2019 ~
  - Staff Engineer, Analog Devices Inc. 2013 ~ 2019
- IEEE **Senior Member**
- IEEE Industrial Electronics Society TC on MEMS/Nanotechnology **Chair** (2019~2020)
- IEEE Sensors Council Technical Achievement Award **Committee** (2020~)
- **Associate Editor**:
  - IEEE Sensors Journal (2017~)
  - IEEE Transactions on Industrial Electronics (2018~2020)
  - IEEE Transactions on Circuit and System – I (2016~2019)

Instructor: Menglian Zhao

- Education:
  - Ph.D.: Zhejiang University
  - M.S.: Zhejiang University
  - B. S.: Hangzhou University
- Working Experience :
  - Associate Professor. Zhejiang University
  - Visiting Scholar, Hongkong University of Science and Technology

Syllabus

- The goal of this course is to **understand integrated circuit**
  - **Read “Paper”**
  - **Think “Circuit”**
  - **Explain “Idea”**

Syllabus

课数	内容	课数	内容
1	导论	9	放大器
2	工艺流程	10	放大器
3	器件模型	11	逻辑门
4	模拟基本单元	12	组合逻辑
5	电流镜与基准	13	时序逻辑
6	单级放大器	14	加法器/乘法器
7	放大器	15	集成电路专题讲座一
8	课堂测验	16	集成电路专题讲座二

Syllabus

- 成绩组合
  - 课堂测验：30%
  - 平时作业：20%
  - 期末考试：50%
- 助教：
  - 杨智迪（yangzd@zju.edu.cn）

Reference Books

- Analog Circuit:
  - **"CMOS Analog Circuit Design" by Phillip Allen**
  - **"Design of Analog CMOS Integrated Circuits" by Behzad Razavi**
- Digital Circuit:
  - **"Digital Integrated Circuits" by Jan Rabaey**

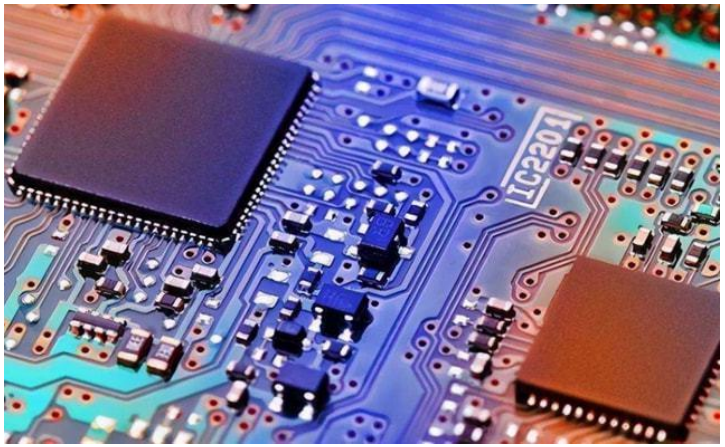
Outline

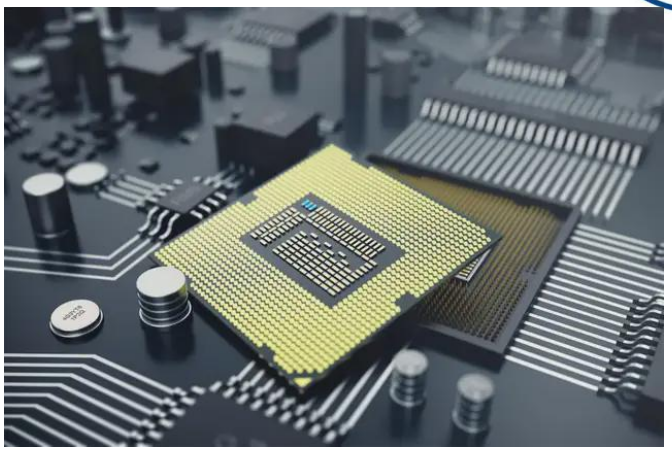
- What is IC and Where it come from:
  - History Class
- Ongoing developments on IC Technology
- Circuits we CARE:
  - Analog Circuits
  - Digital Circuits
- New Era for IC researcher

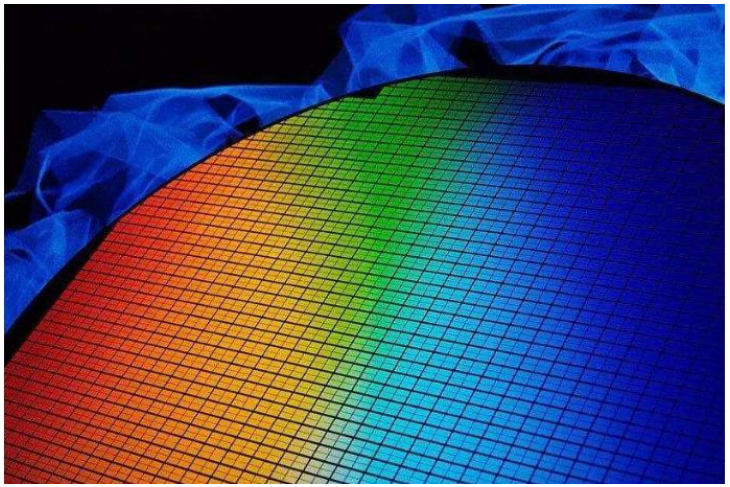
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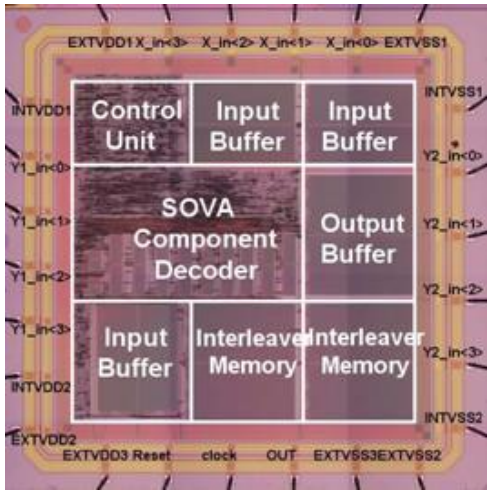
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IC: Integrated Circuit

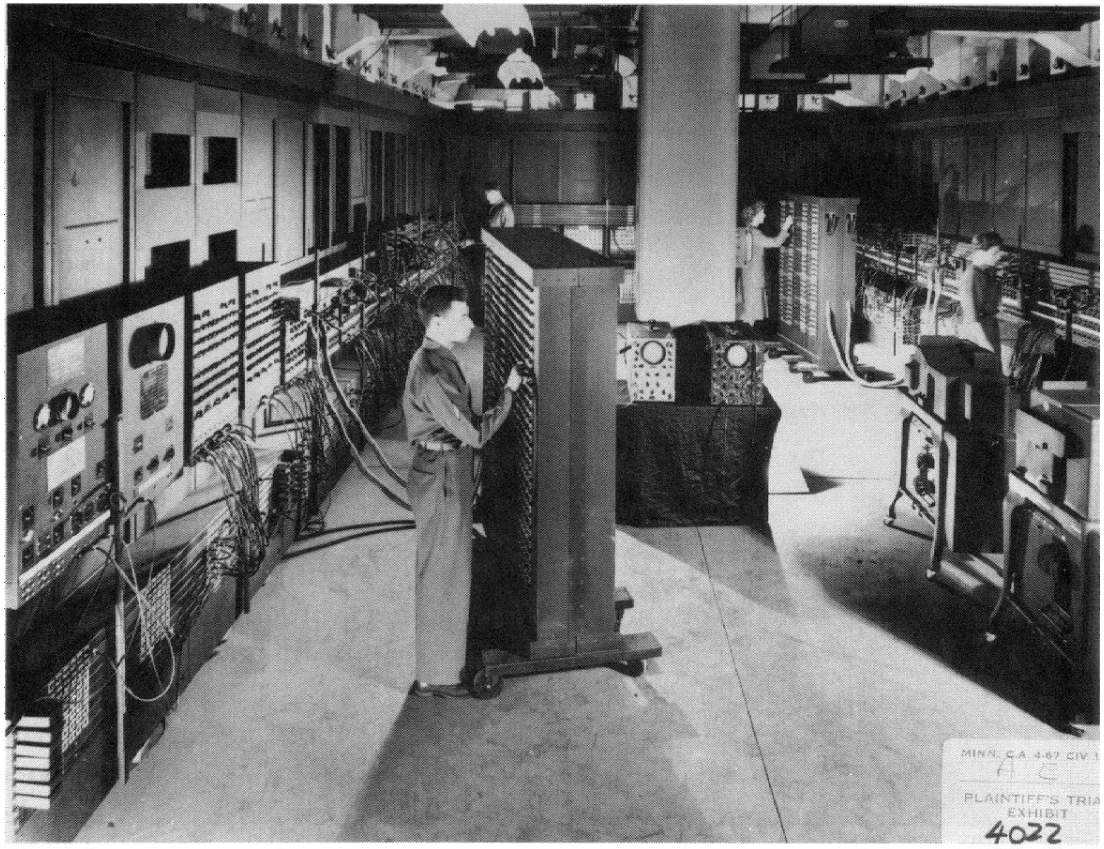








Vacuum Tubes

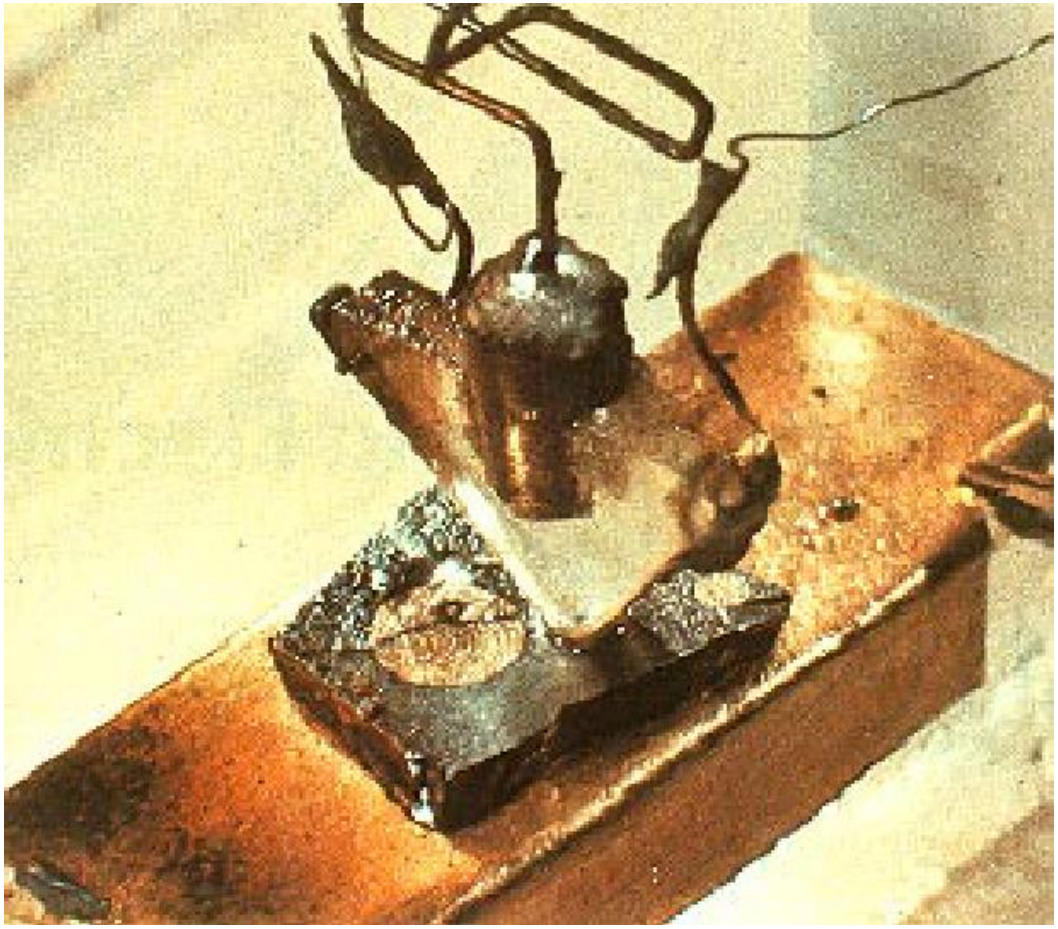


ENIAC:

Electronic Numerical Integrator and Calculator

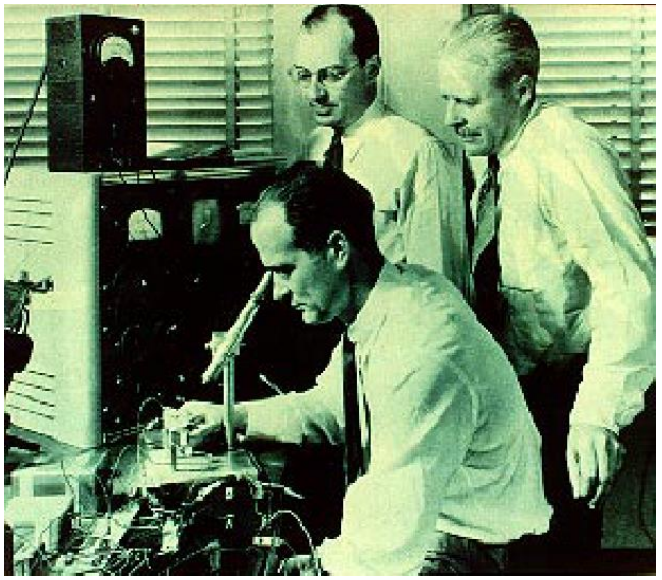
- The 1st General-purpose Electronic computer
- February 14rd, 1946
- Moore School, Univ. of Pennsylvania
- 18,000 vacuum tubes
- 5000ADD/sec
- 30 Ton
- 140kW
- MTBF: 7min

The First: Transistor



On December 23rd, 1947, the semiconductor research team of Bell laboratory had invented the first NPN Ge transistor

1949 Schockley Bipolar transistor



Shockley Lab

- **William Shockley**



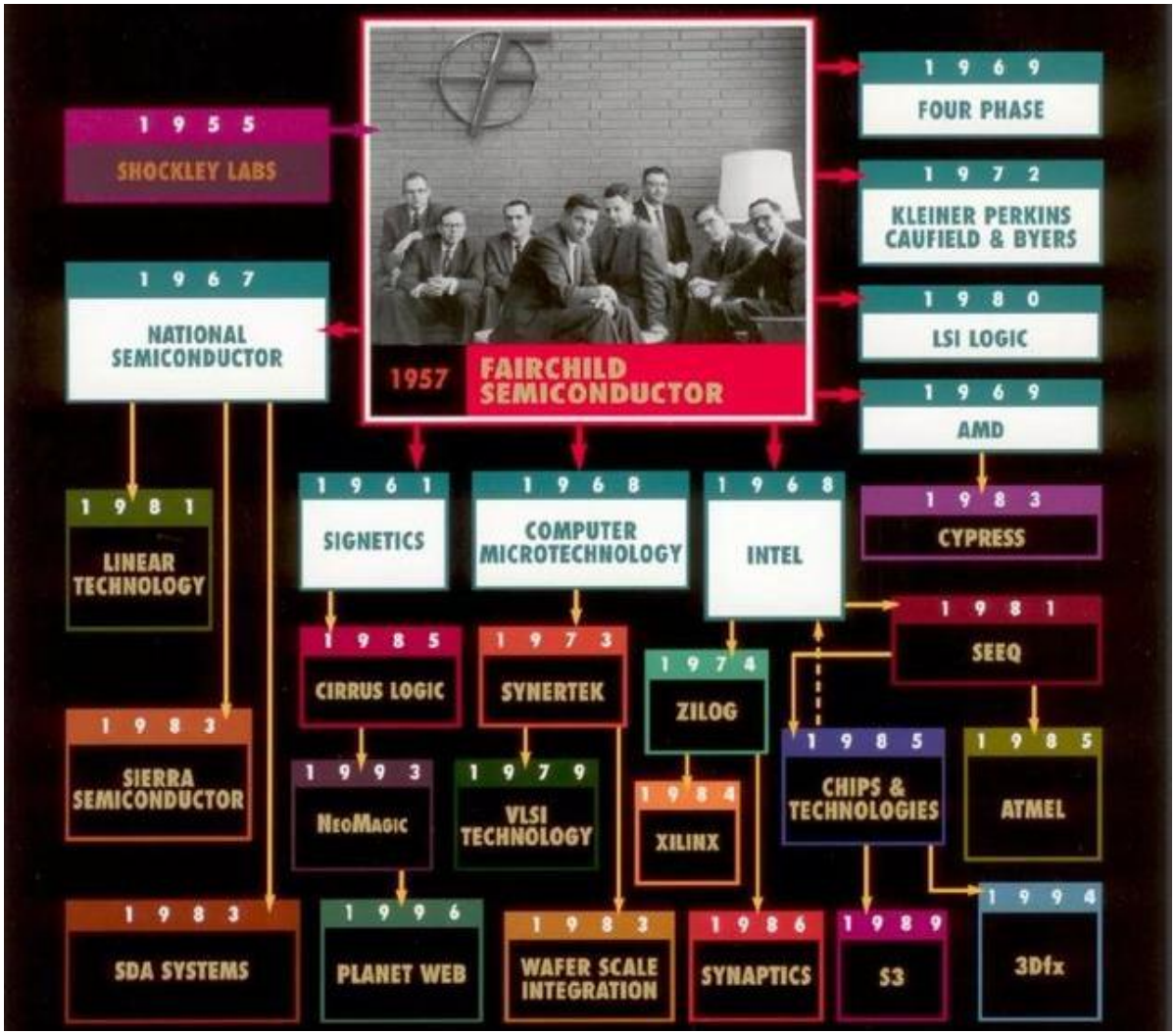
1956 Nobel prize in Physics

Traitorous 8



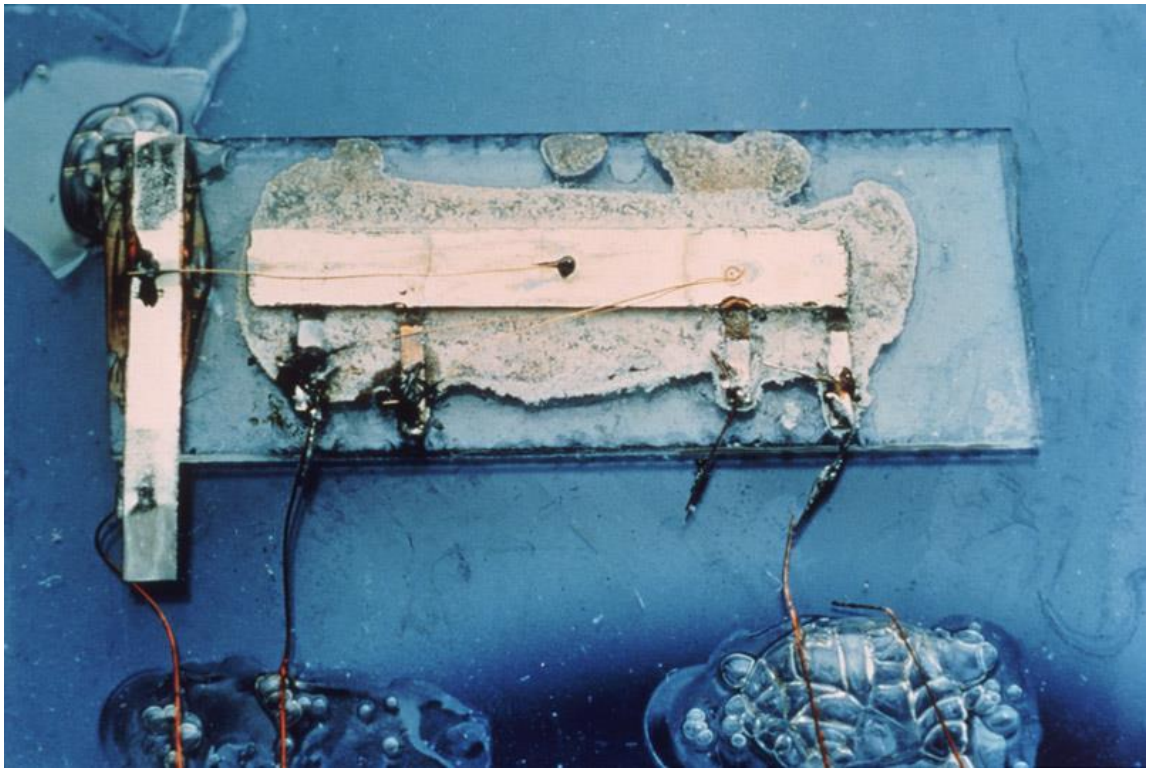
- Fairchild Semiconductor(1959): **Gordon Moore**, Sheldon Roberts, Eugene Kleiner, **Robert Noyce**, Victor Grinich, Julius Blank, Jean Hoerni , Jay Last

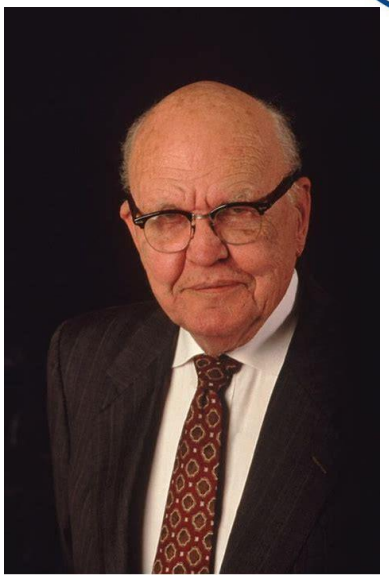
The Seed: Fairchild Semi.



The 92 public companies that can be traced back to Fairchild are now worth about $2.1 trillion

The First Integrated Circuit

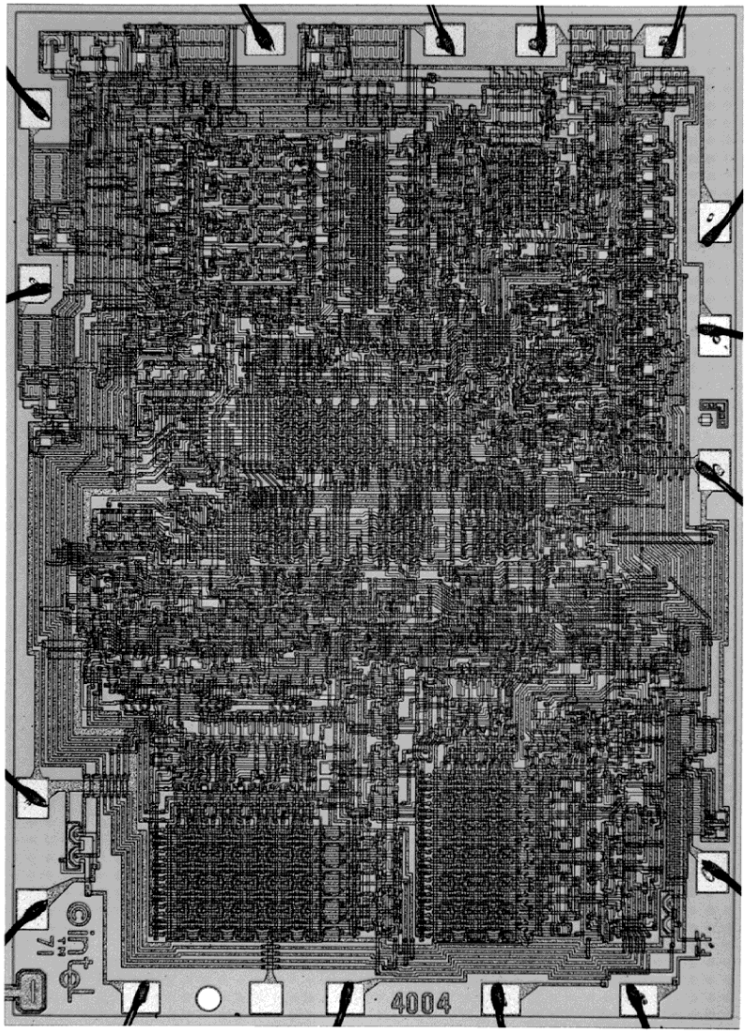




In 1958, Jack Kilby and his research team in TI Corporation had invented the first IC: 12 Component, Ge semiconductor.

2000 Nobel Prize in Physics

Intel 4004 Micro-Processor



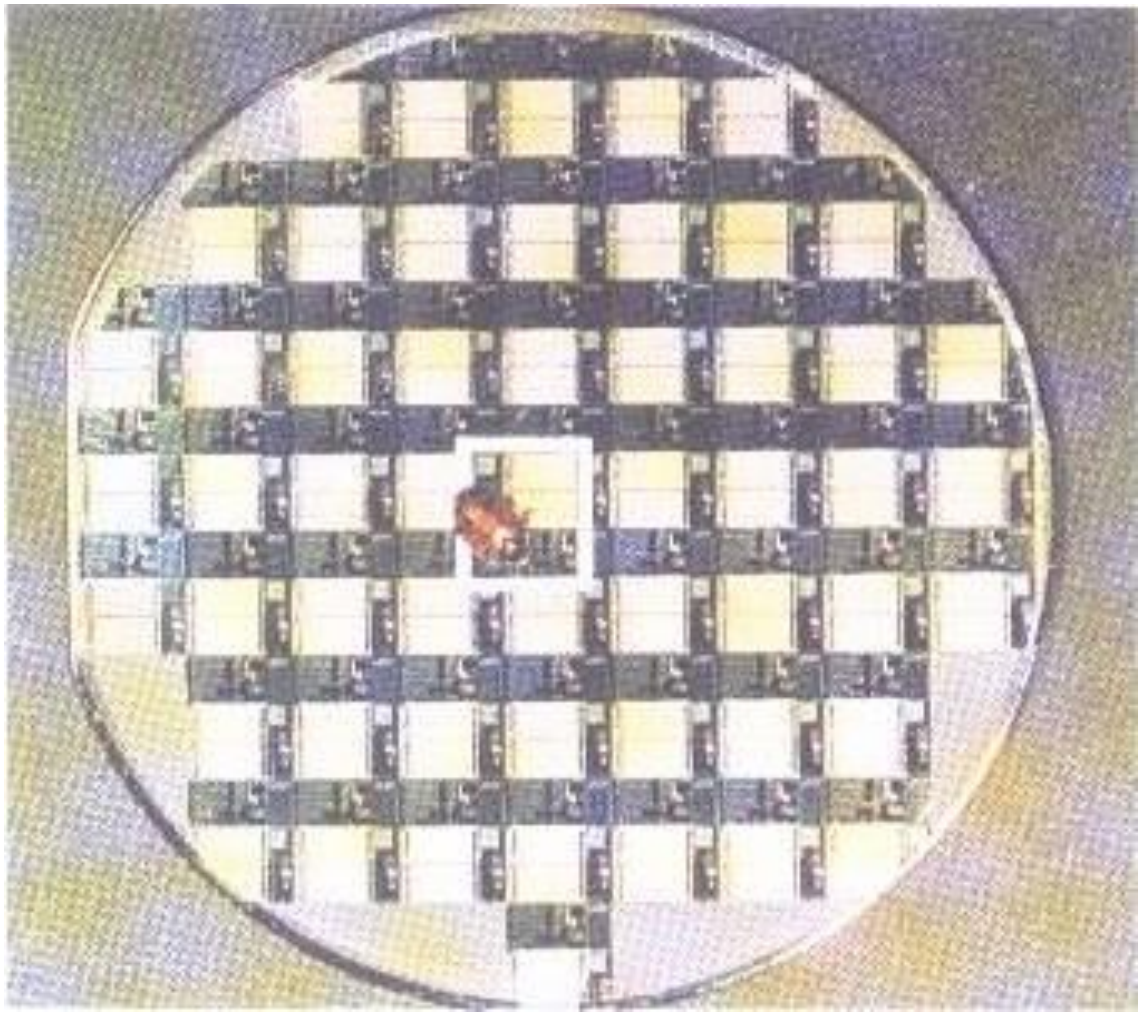
First Microprocessor 4004

1971-72

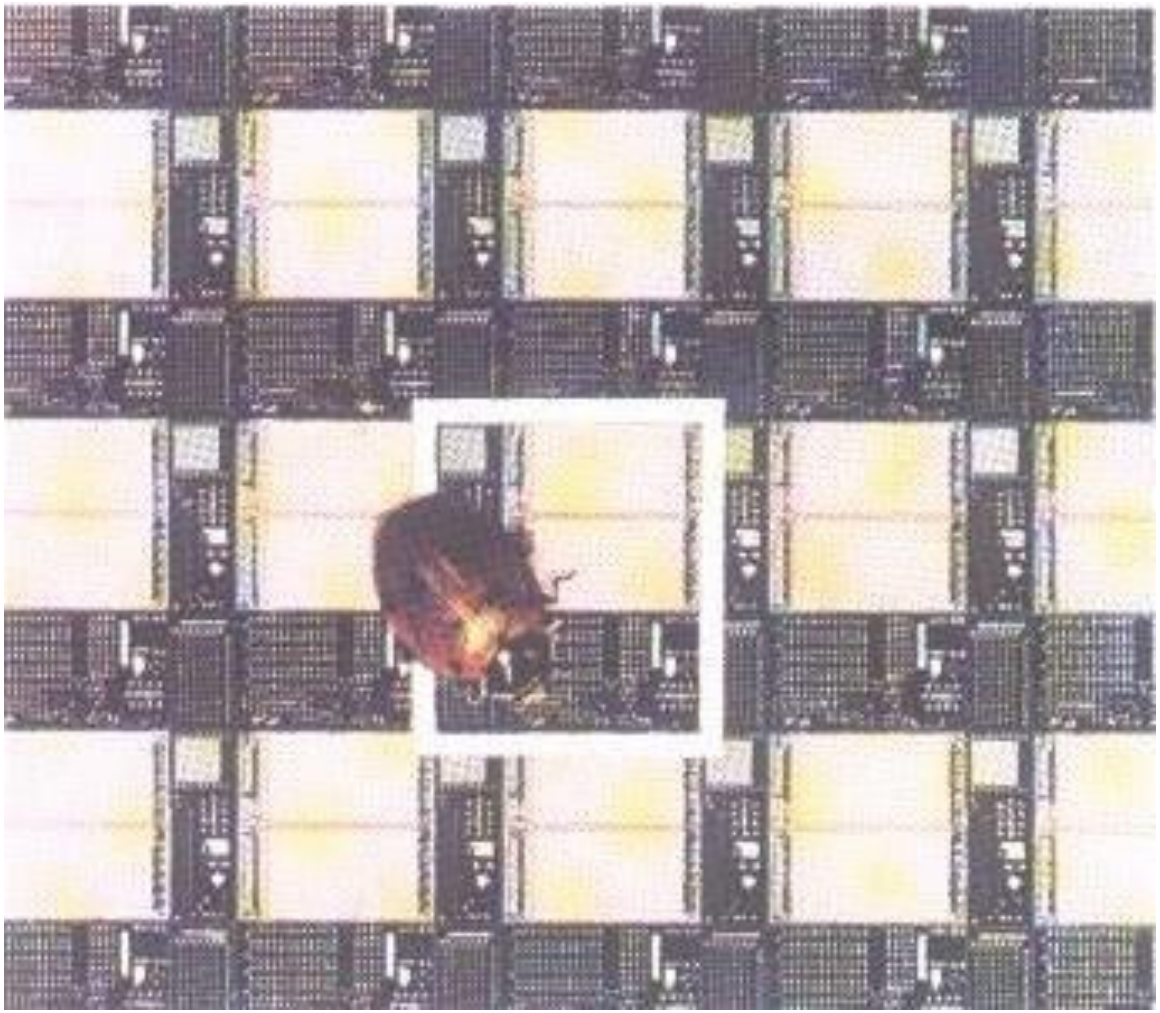
1000 transistors

1 MHz operation

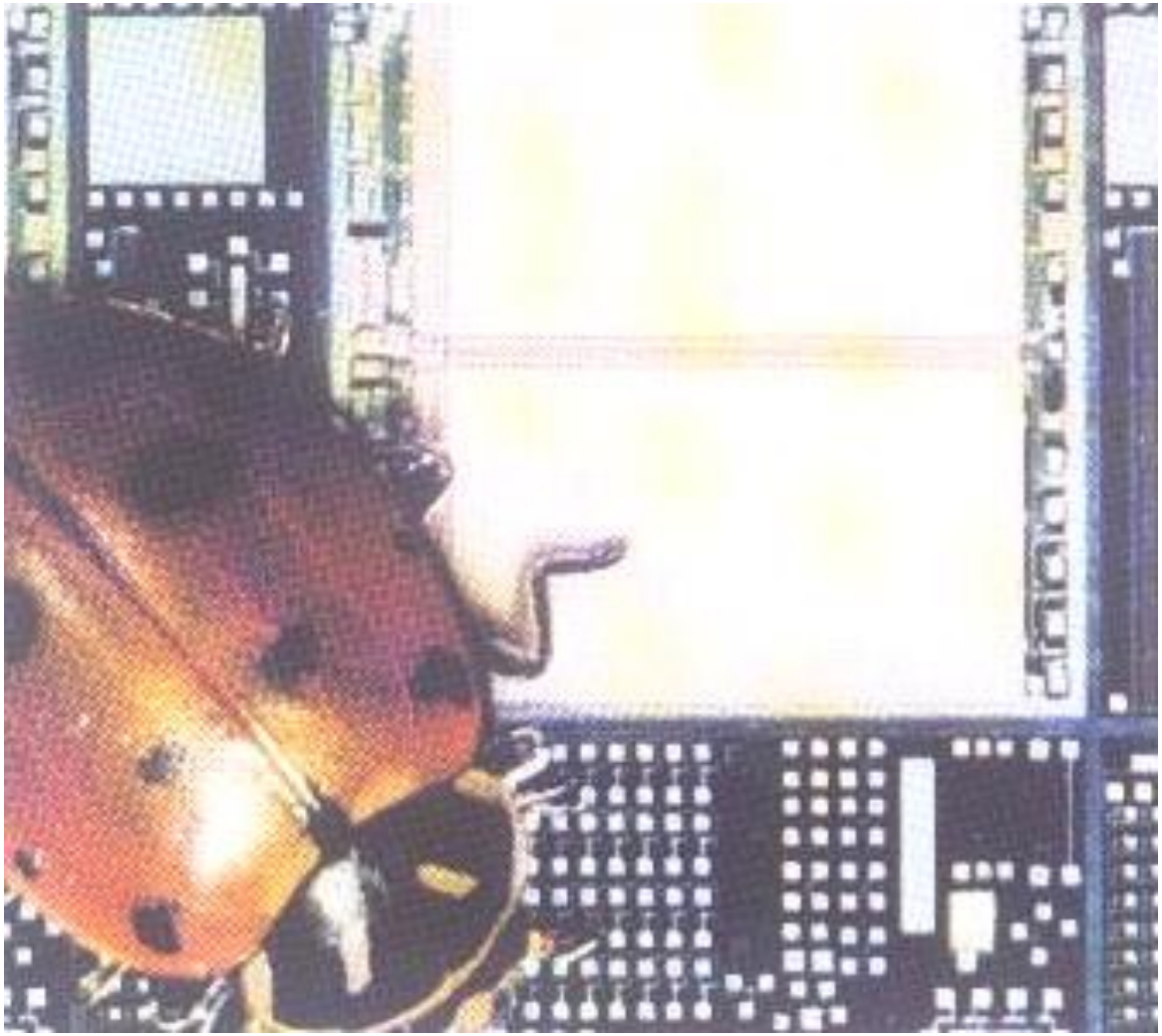
Integrated



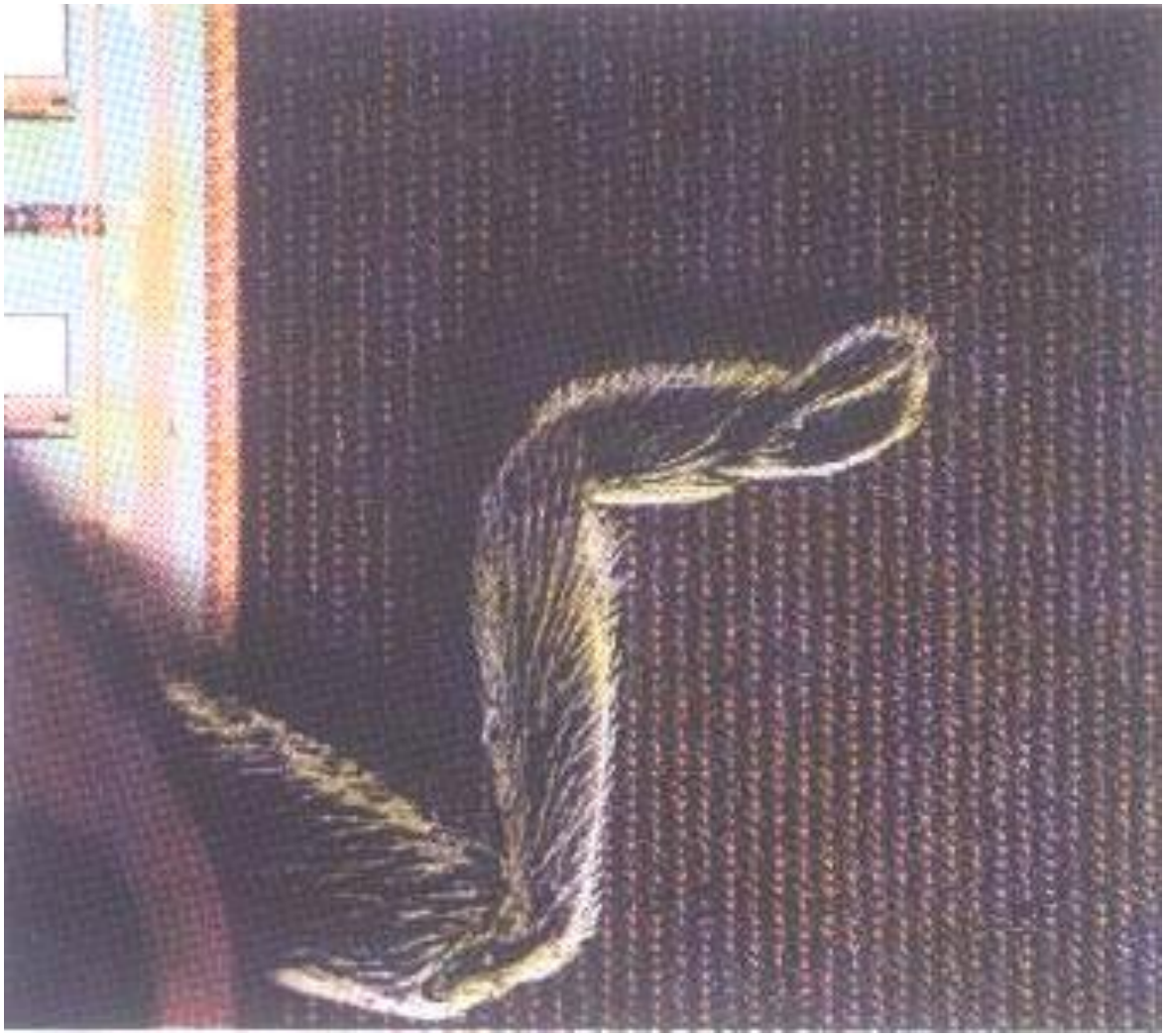
Integrated



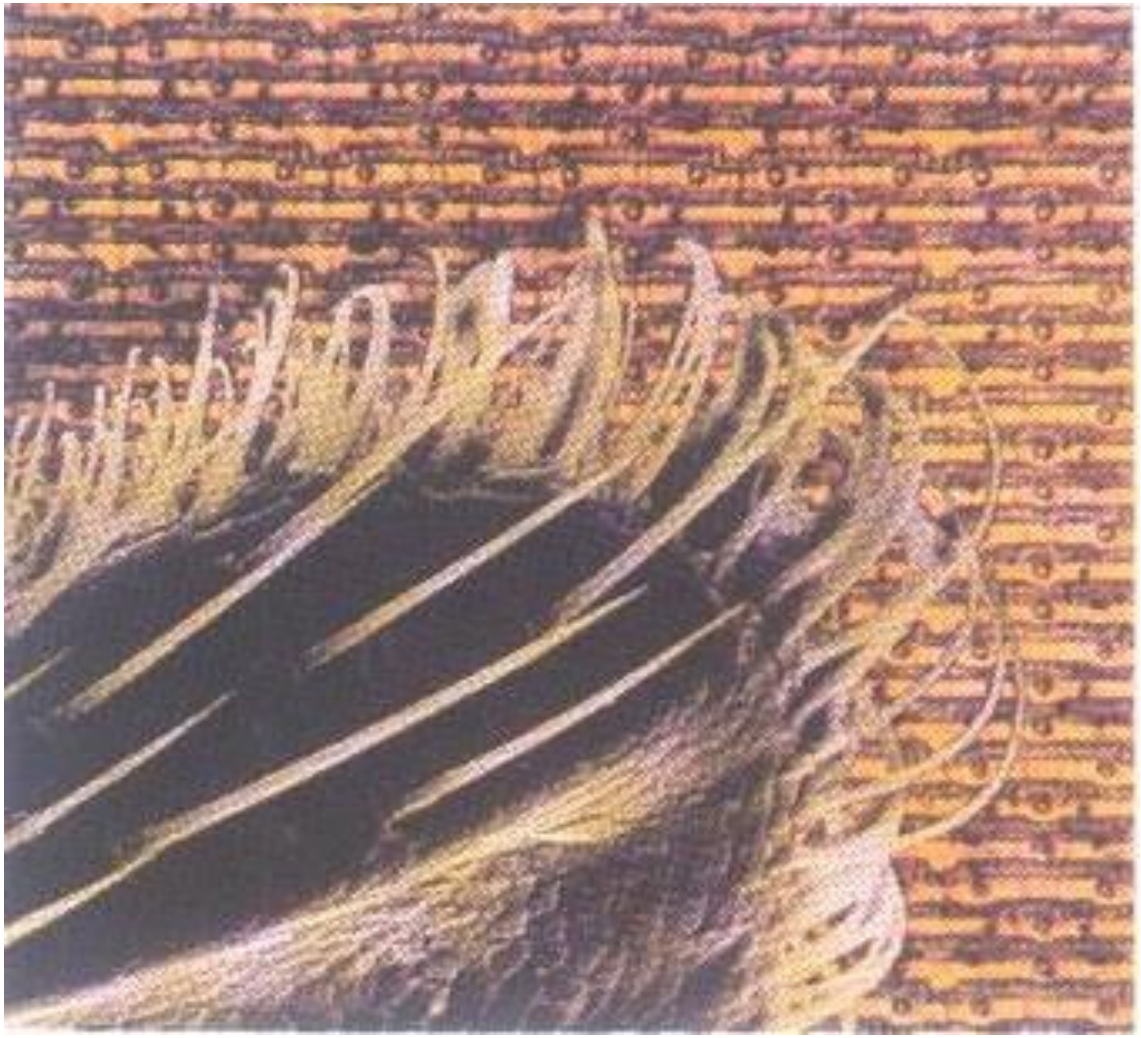
Integrated



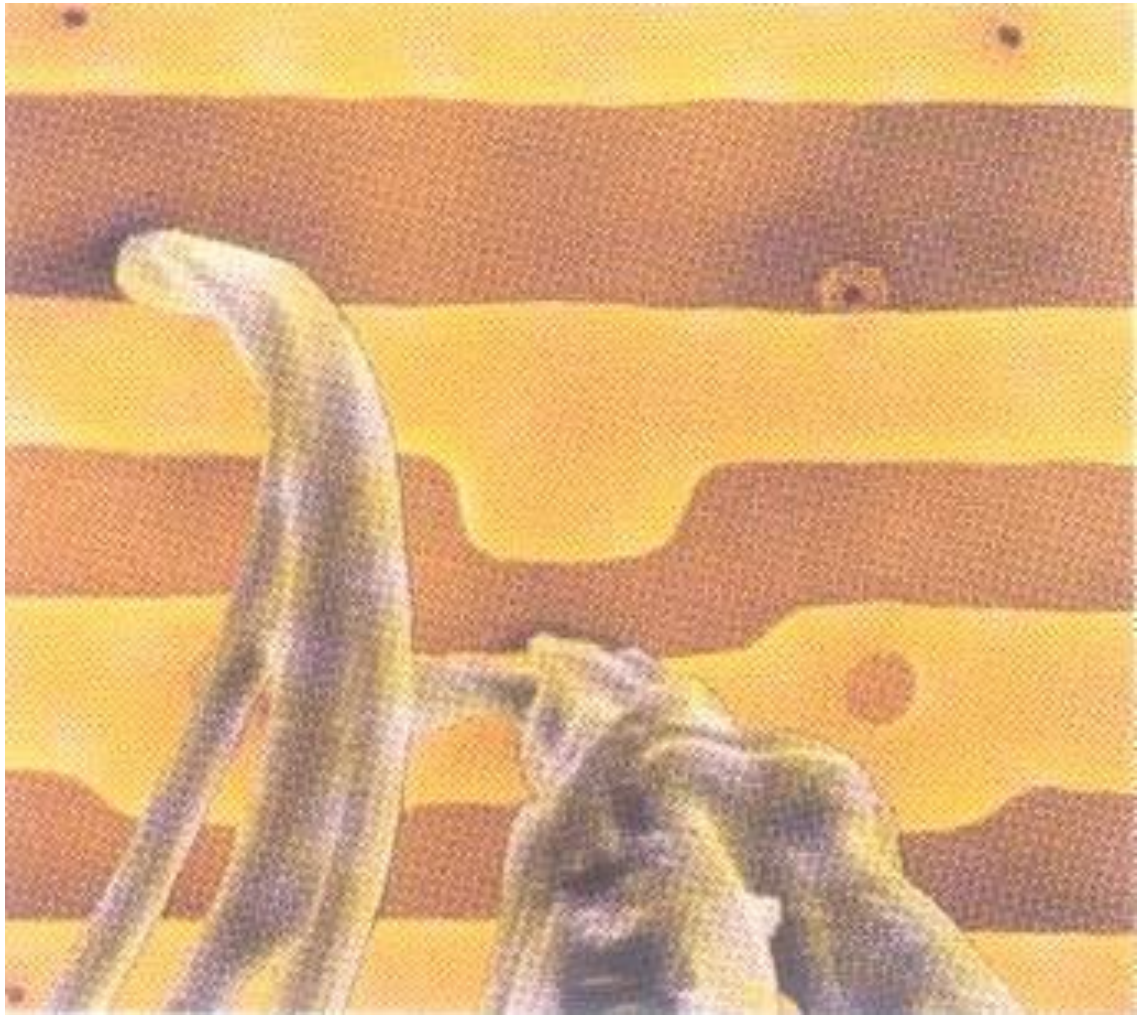
Integrated



Integrated



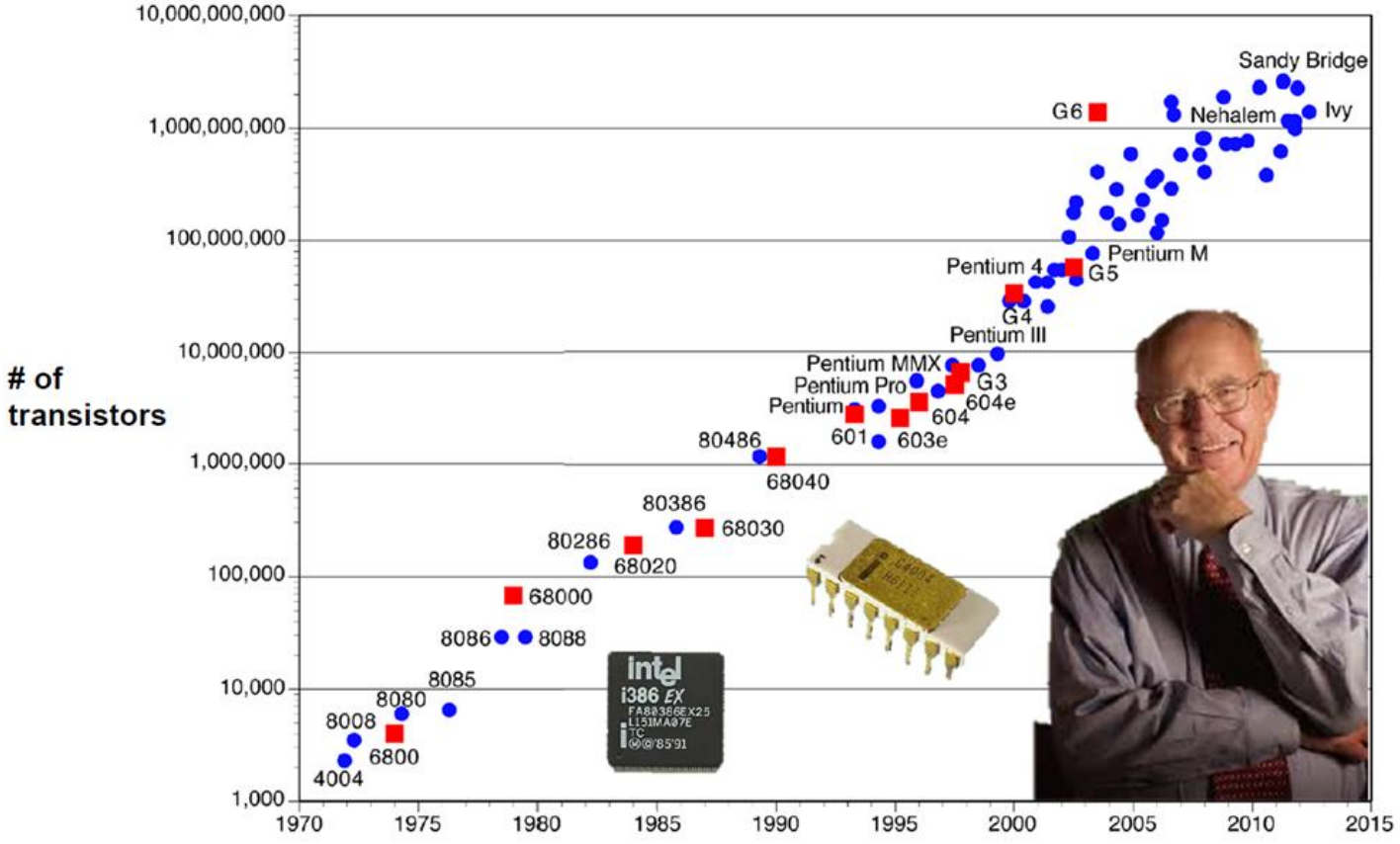
Integrated



Outline

- What is IC and Where it come from:
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- New Era for IC researcher

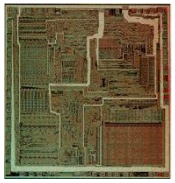
Moore's Law

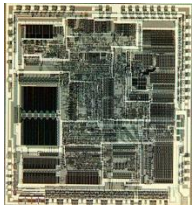


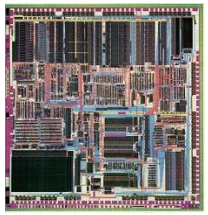
- The observation that the number of transistors in a dense IC doubles approximately every two years

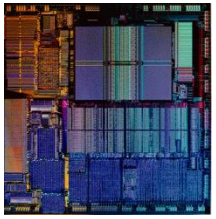
Processor

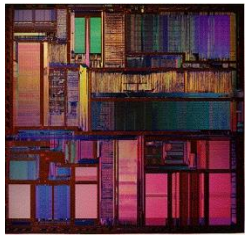

8080

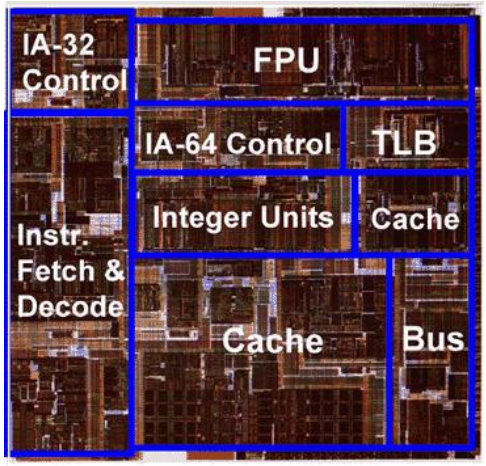

8086


80286

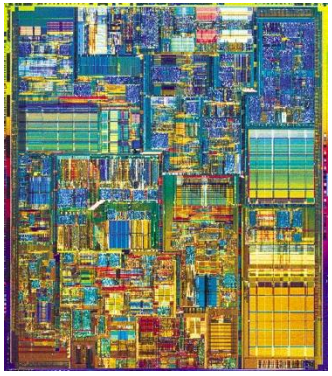

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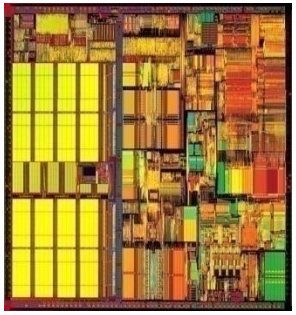

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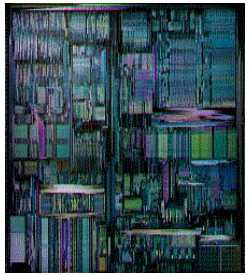

Pentium



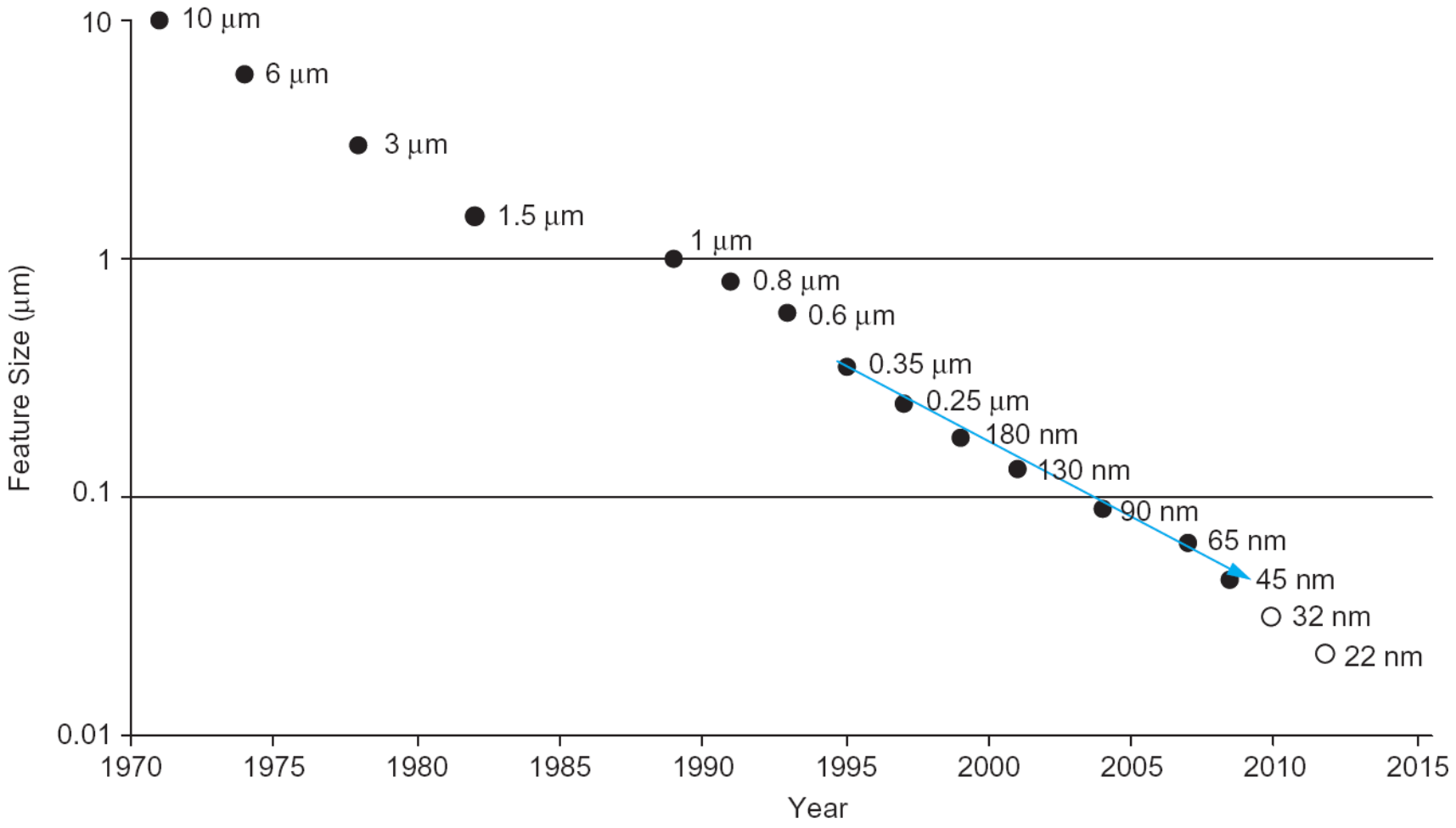
Itanium


Pentium IV


Pentium III


PentiumPro

Process Generations



From PC Age to Internet Age

Stationary PC World → **Dynamic, Real Time Internet Age**

PC Age

Computer Focus



- Non-Real Time
- Stationary
- Digital I/O
- µP and Memory

Internet Age

Communication Focus

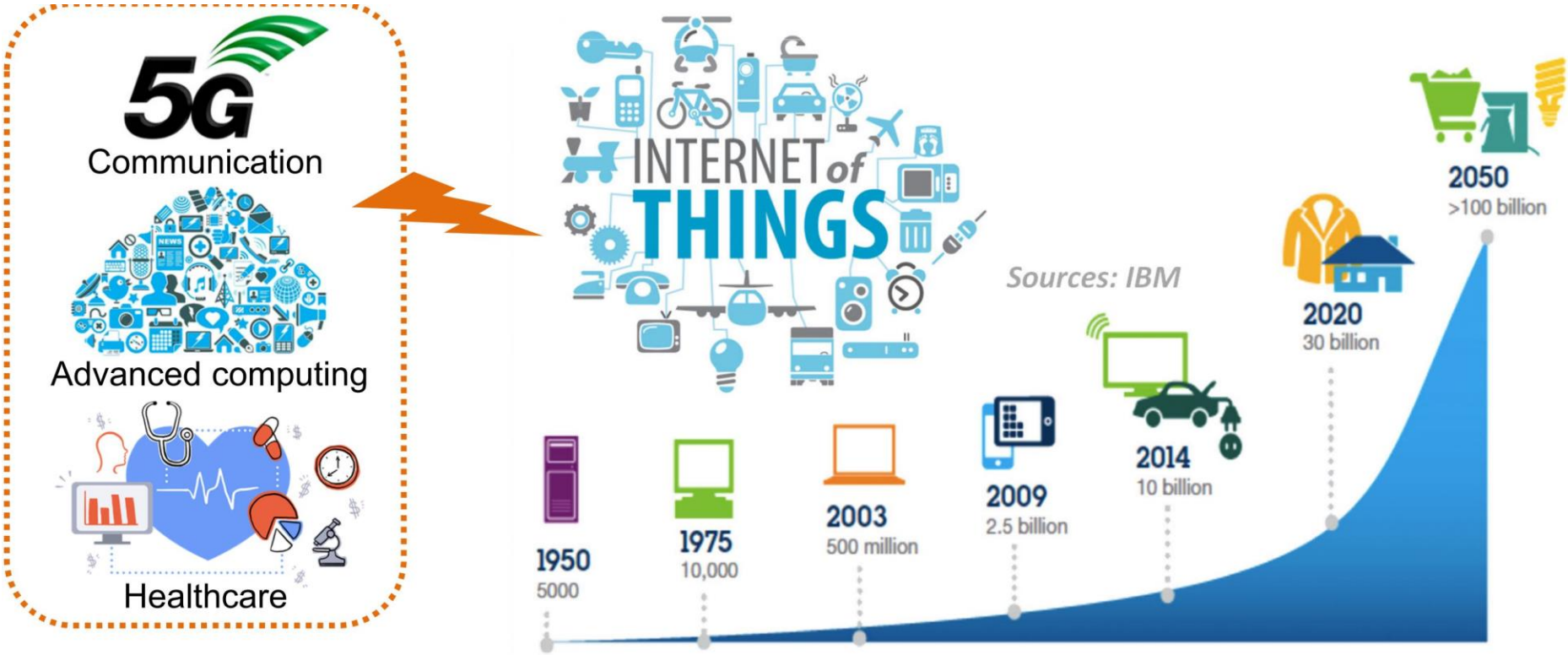


- Real Time
- Mobile
- Analog I/O
- DSP and Analog

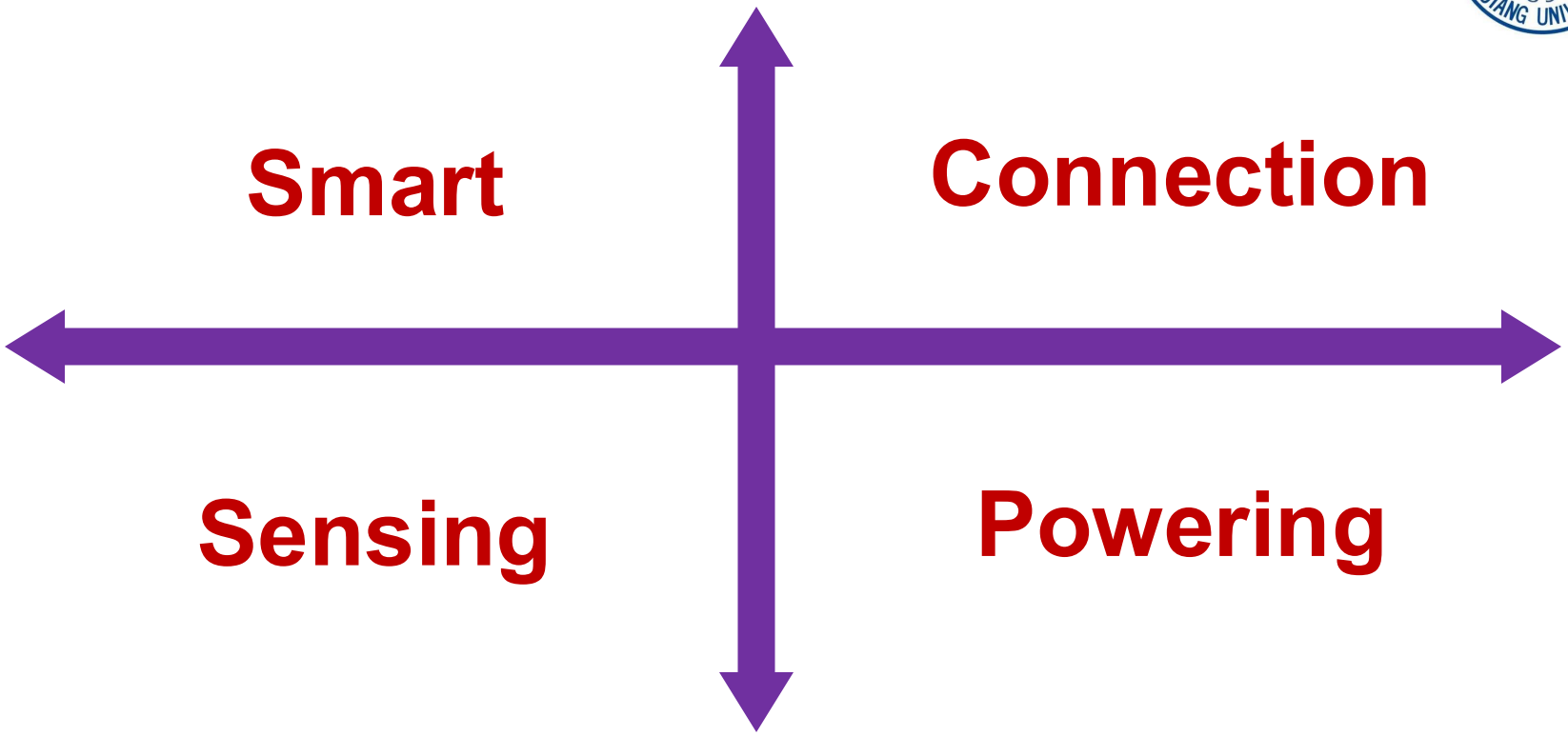


Source: Dennis Buss, TI

From Internet Age to IoT Era



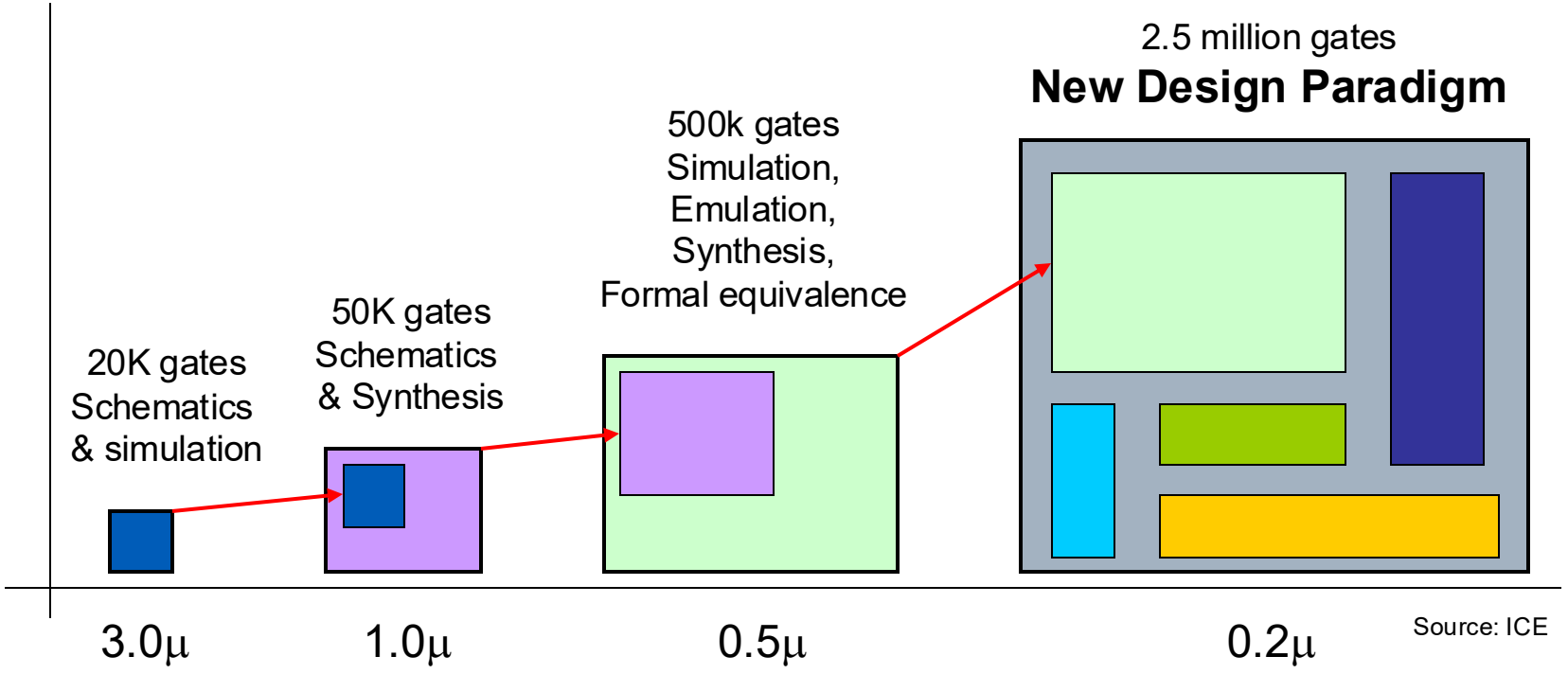
Opportunity in IoT Era: Set and Forget



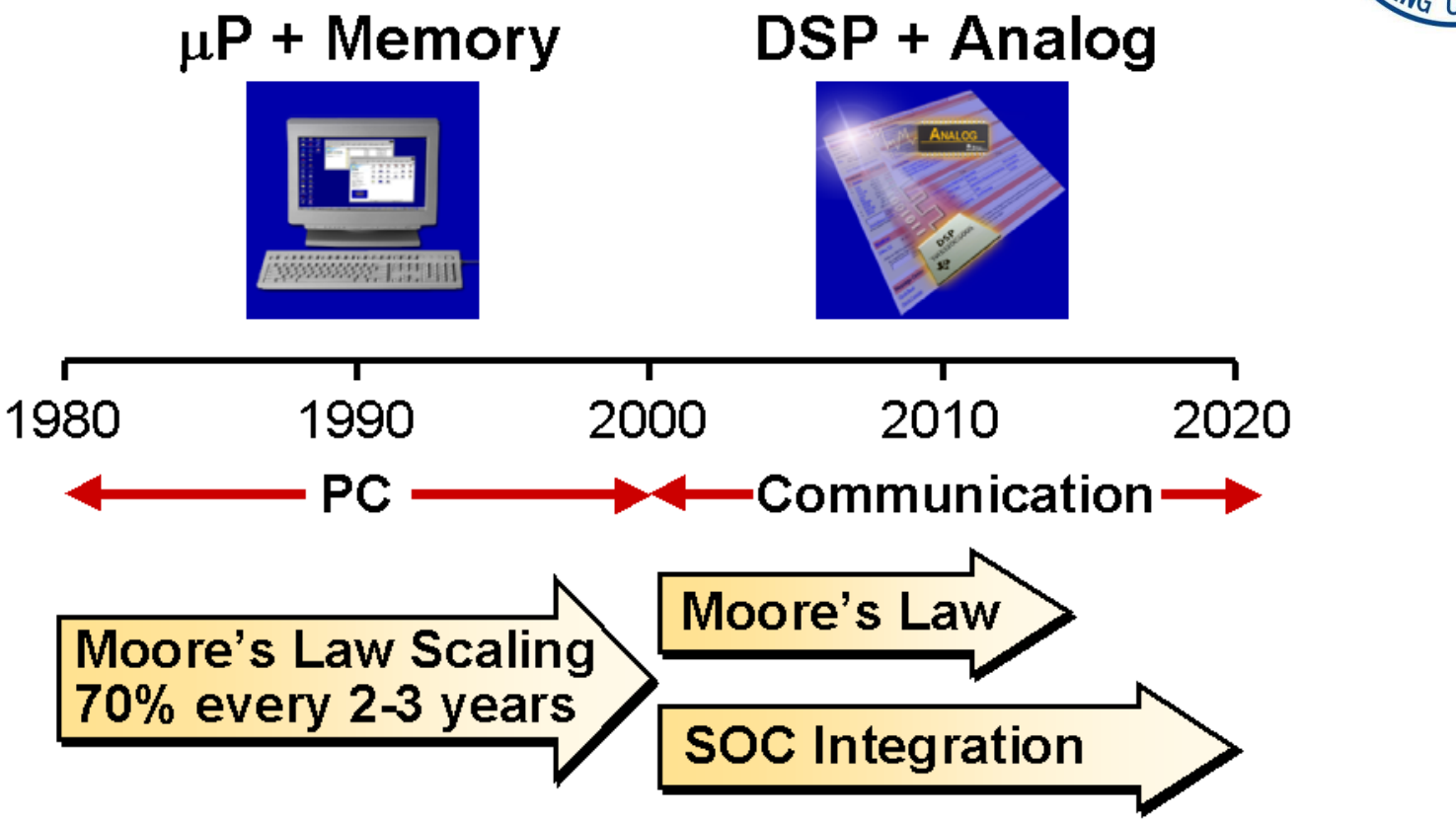
SOC: System on Chip

System on a Chip development:

Yesterday's chip is today's functional block!



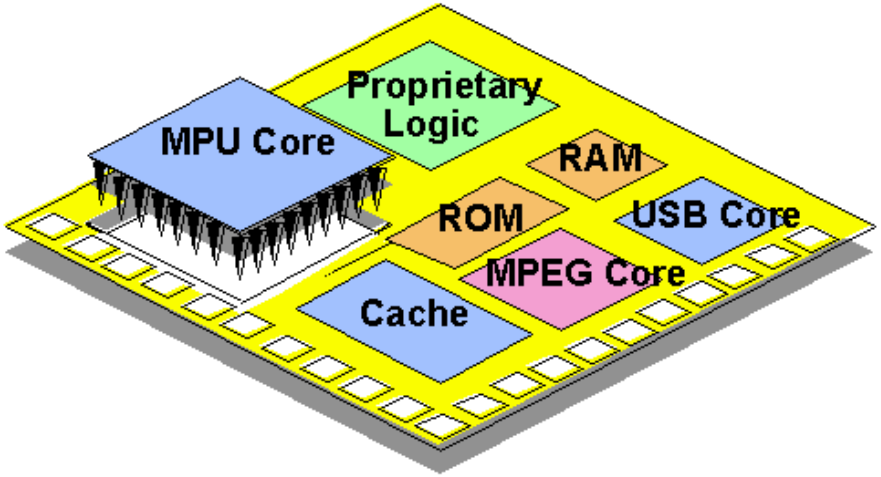
SOC: System on Chip



Transistor scaling will continue to be an important technology driver. But it will no longer be the sole driver – SOC integration will accelerate in the Internet age.

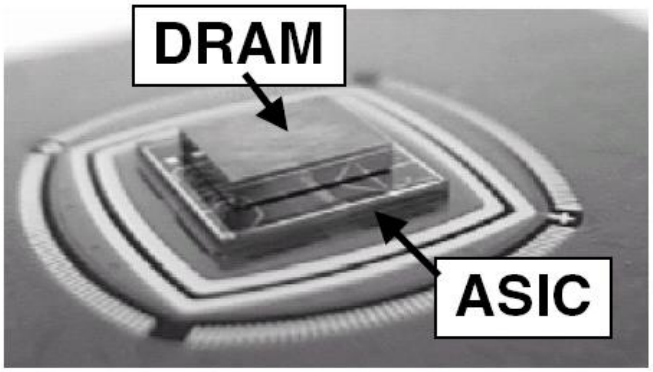
Integration!

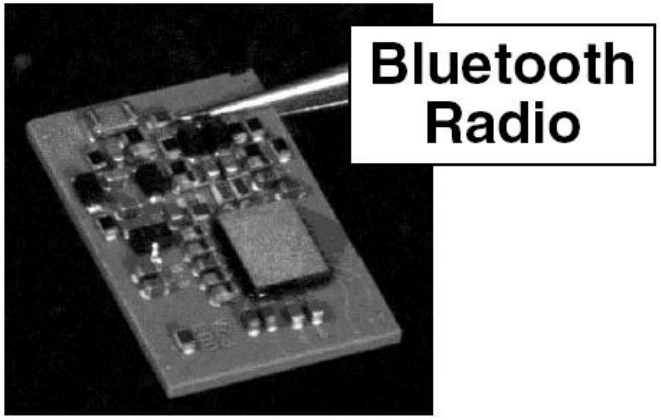
- **Processor Cores**
  - CPU, DSP
- **Memories**
  - RAM (DRAM, SRAM)
  - ROM
- **Reconfigurable Comp**.
  - field-programmable logic (FPGA)
  - configurable interconnect

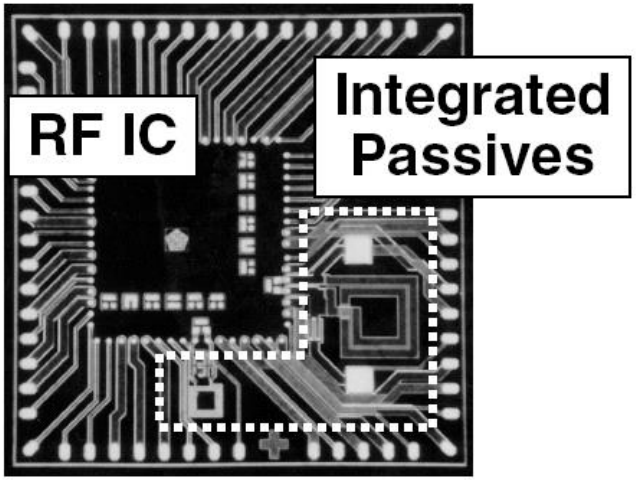


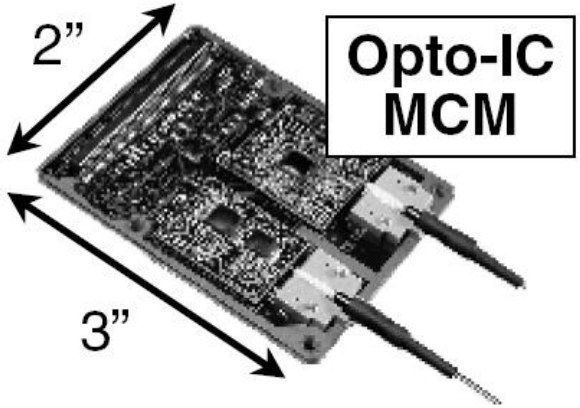
- **Micro-Electro-Mechanical Systems (MEMS) - Microsystem Comp.**
  - low cost & increased performance
  - optical MEMS
- **Analog RF Components, Mixed-Signal Components**
  - LNA, VCO, mixers, inductors
  - ADCs & DACs
- **Software for SoCs**

SIP: System In Package





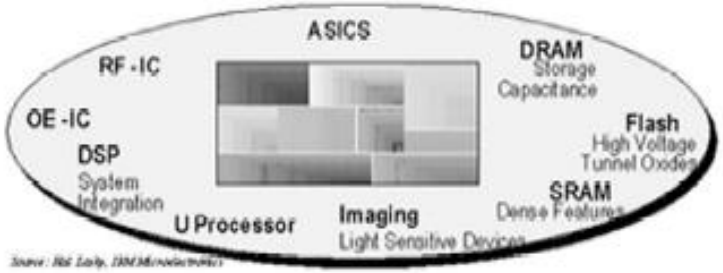




SIP: System In Package

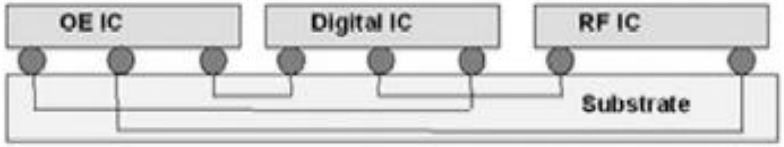
SOC

a) Complete system on one chip



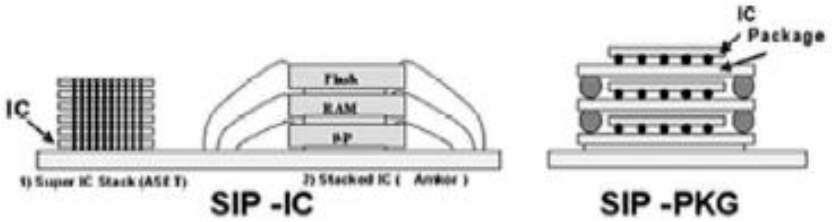
MCM

b) Interconnects components



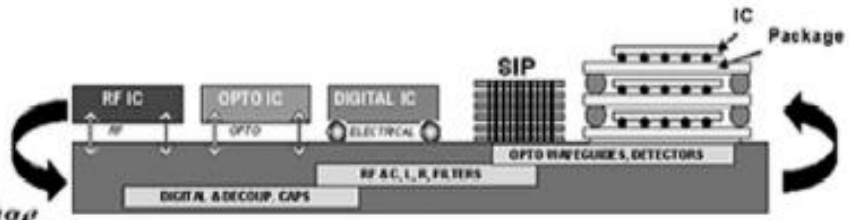
SIP

c) Stackeds chips or packages for reduced form factors

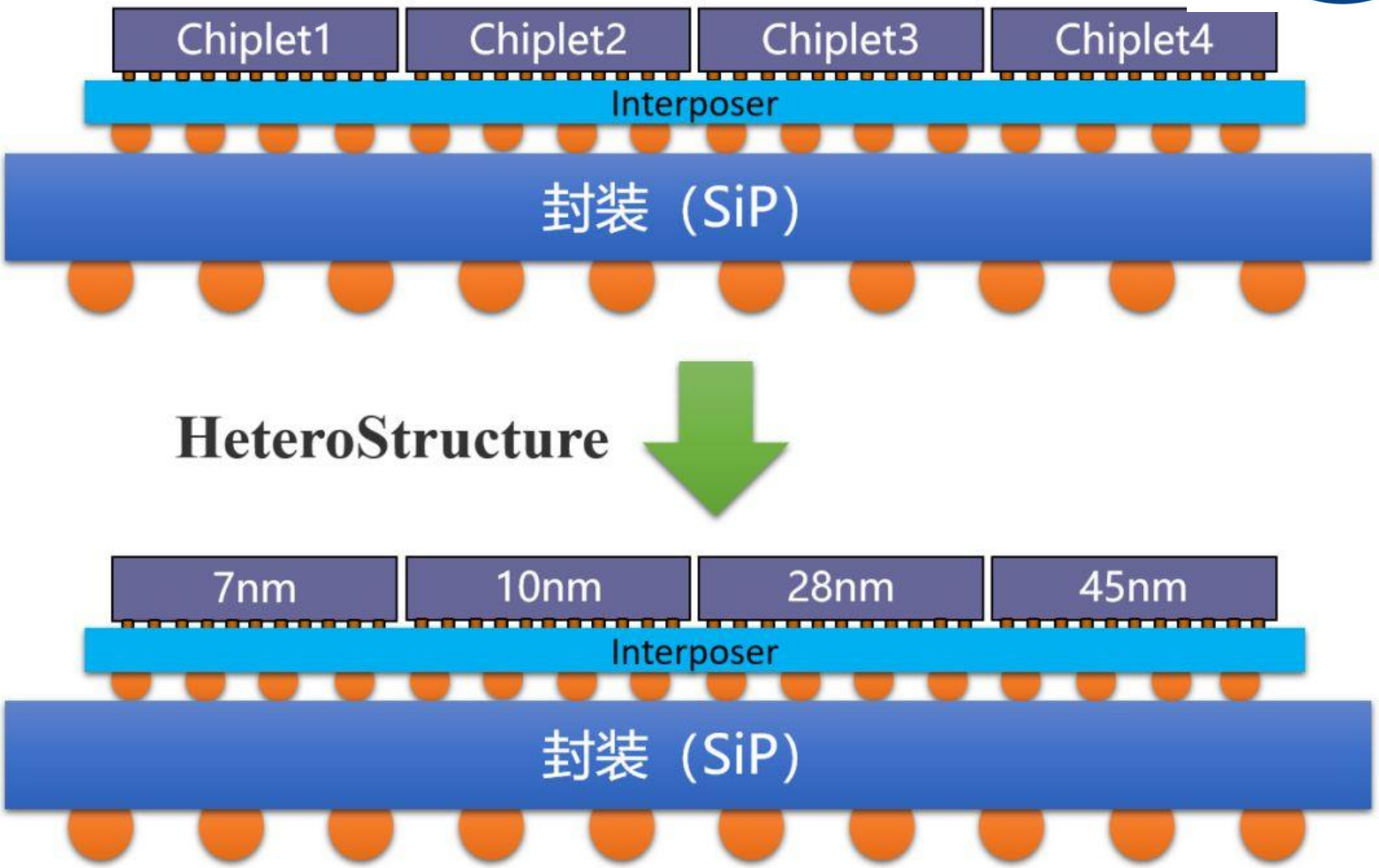


SOP

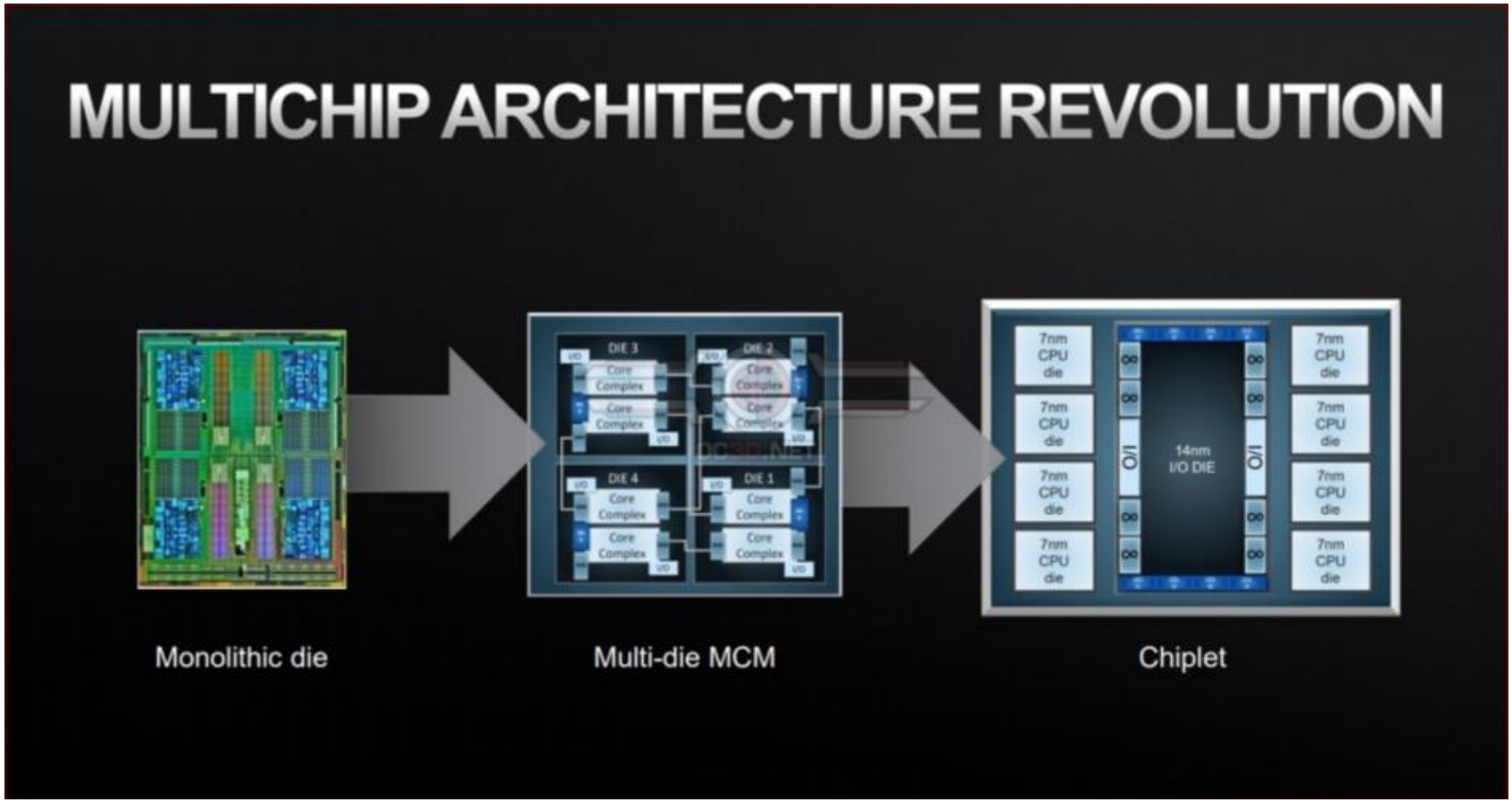
d) Optimizes functions between ICs and package. Miniaturizes systems.



Challenge the Law: CHIPLET



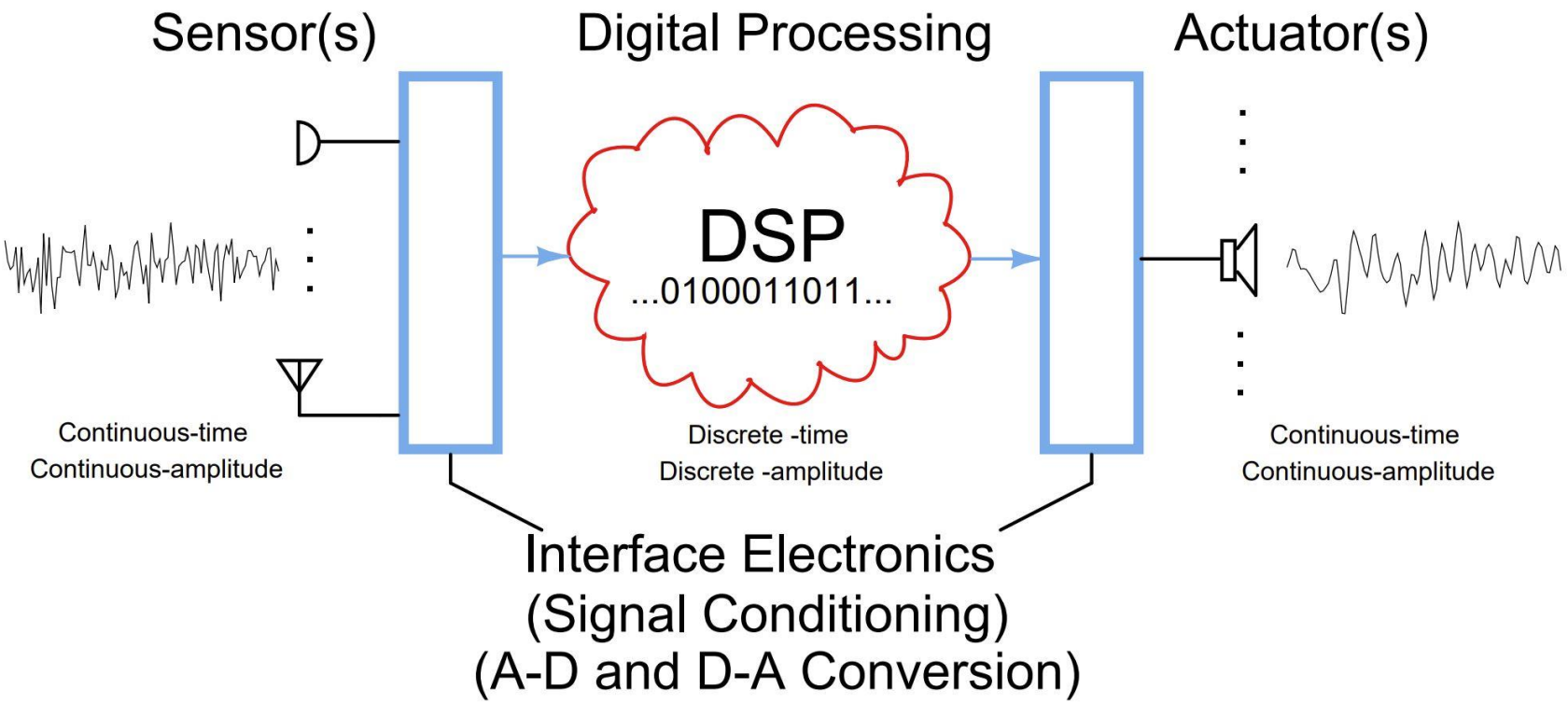
Challenge the Law: CHIPLET



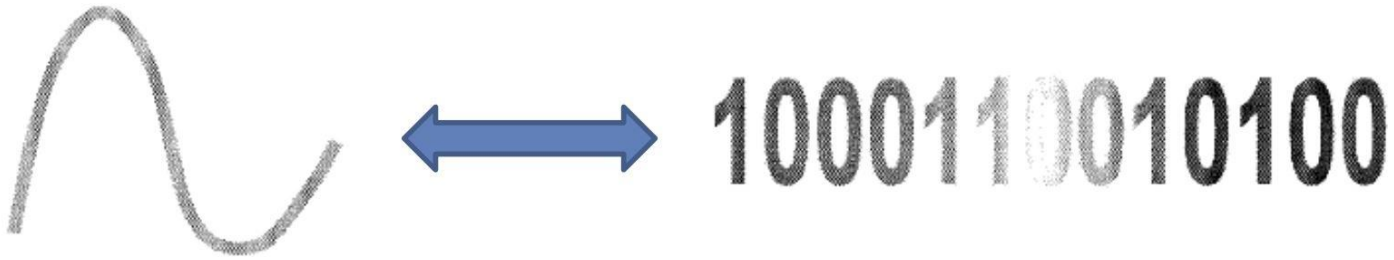
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Mixed Signal (A-D and D-A)



Analog and Digital



Analog signal:

Time continuous

Amplitude continuous

Physical quantity

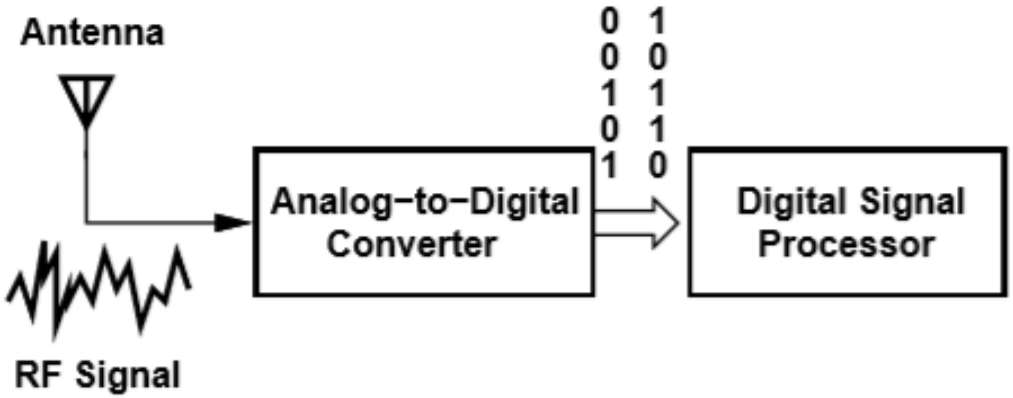
Digital signal:

Time discrete

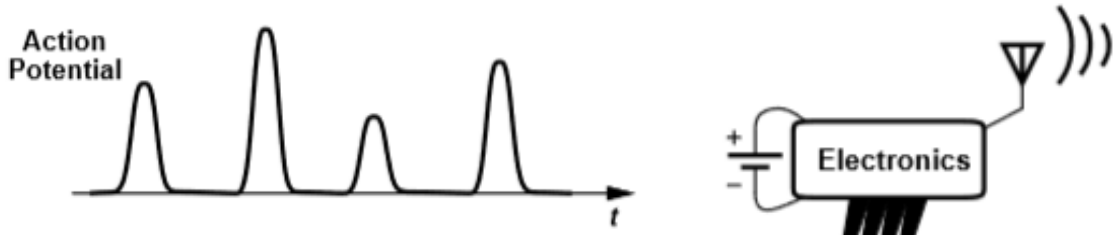
Amplitude discrete

Numerical value

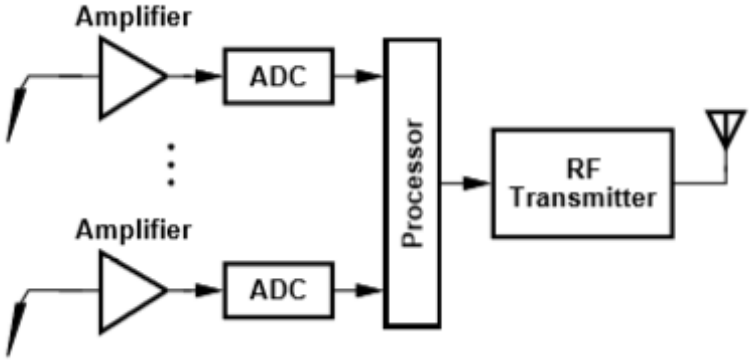
Why Analog?



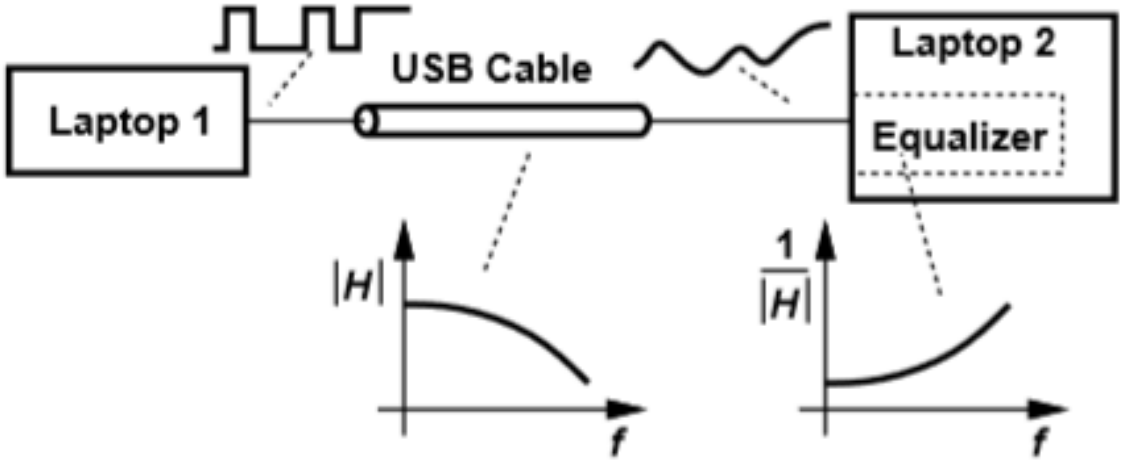
RF System



Neural System



When Digital become Analog



- Equalization to compensate high-frequency attenuation in a USB cable.

How Fast of Analog Signal

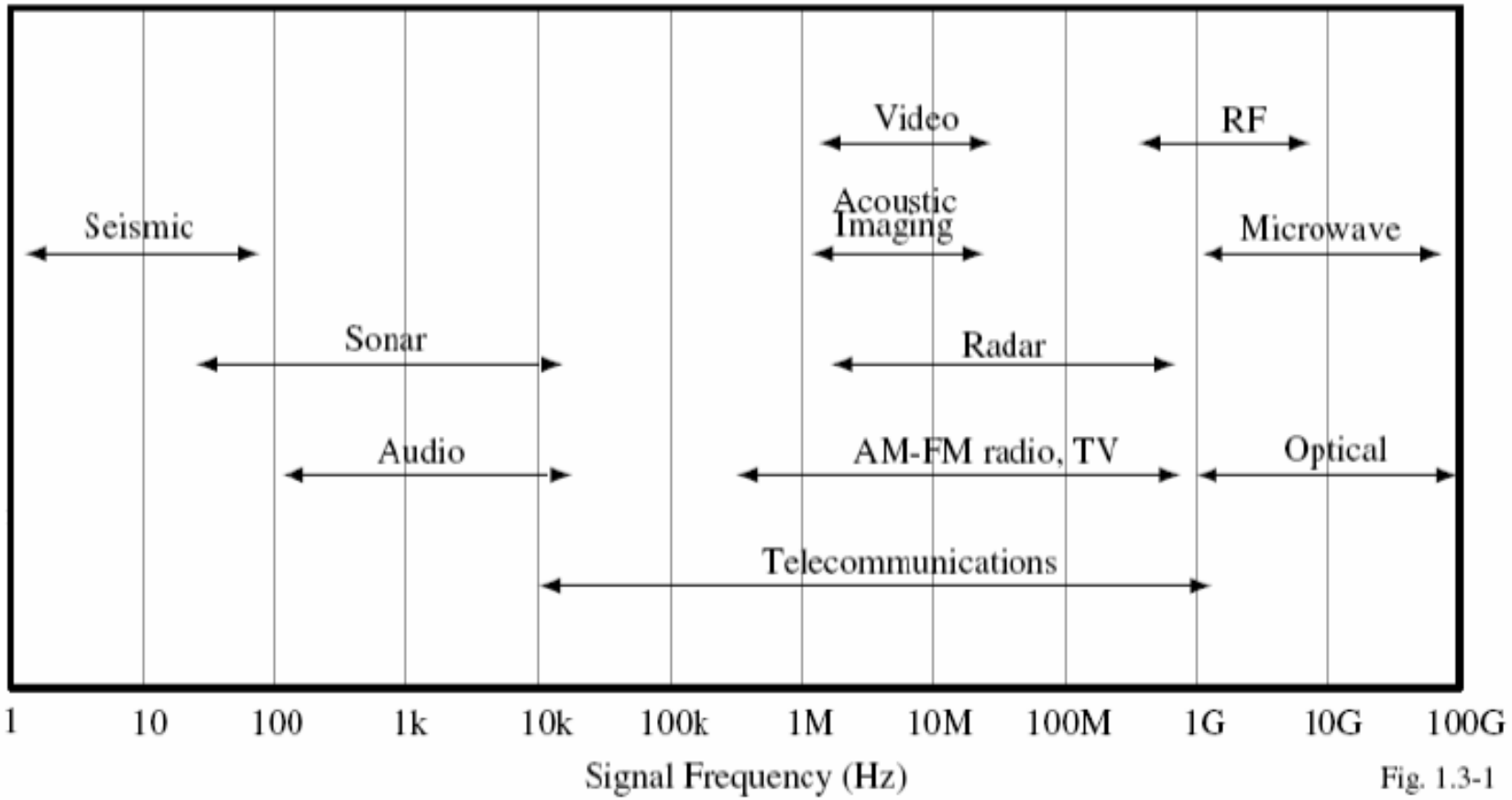
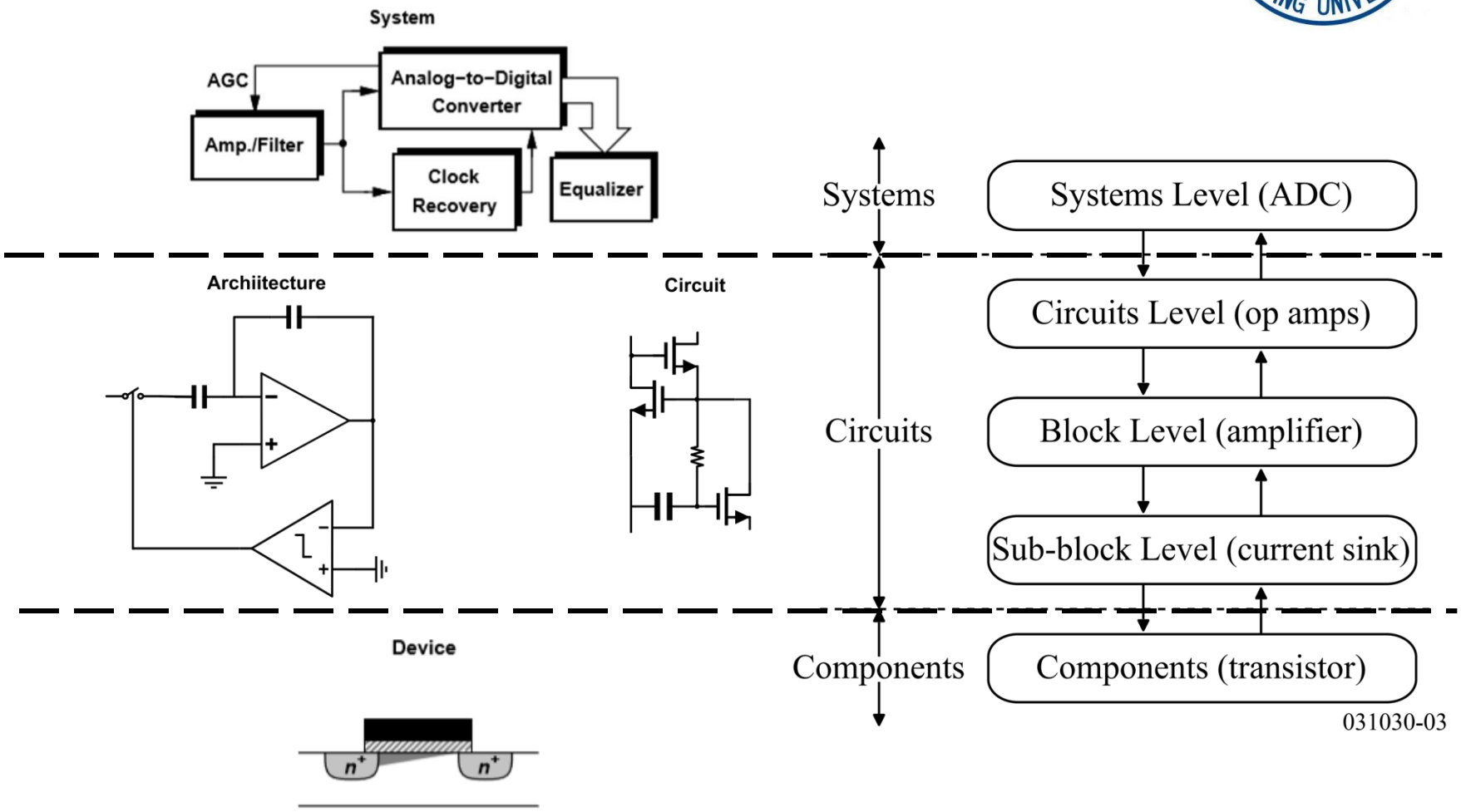
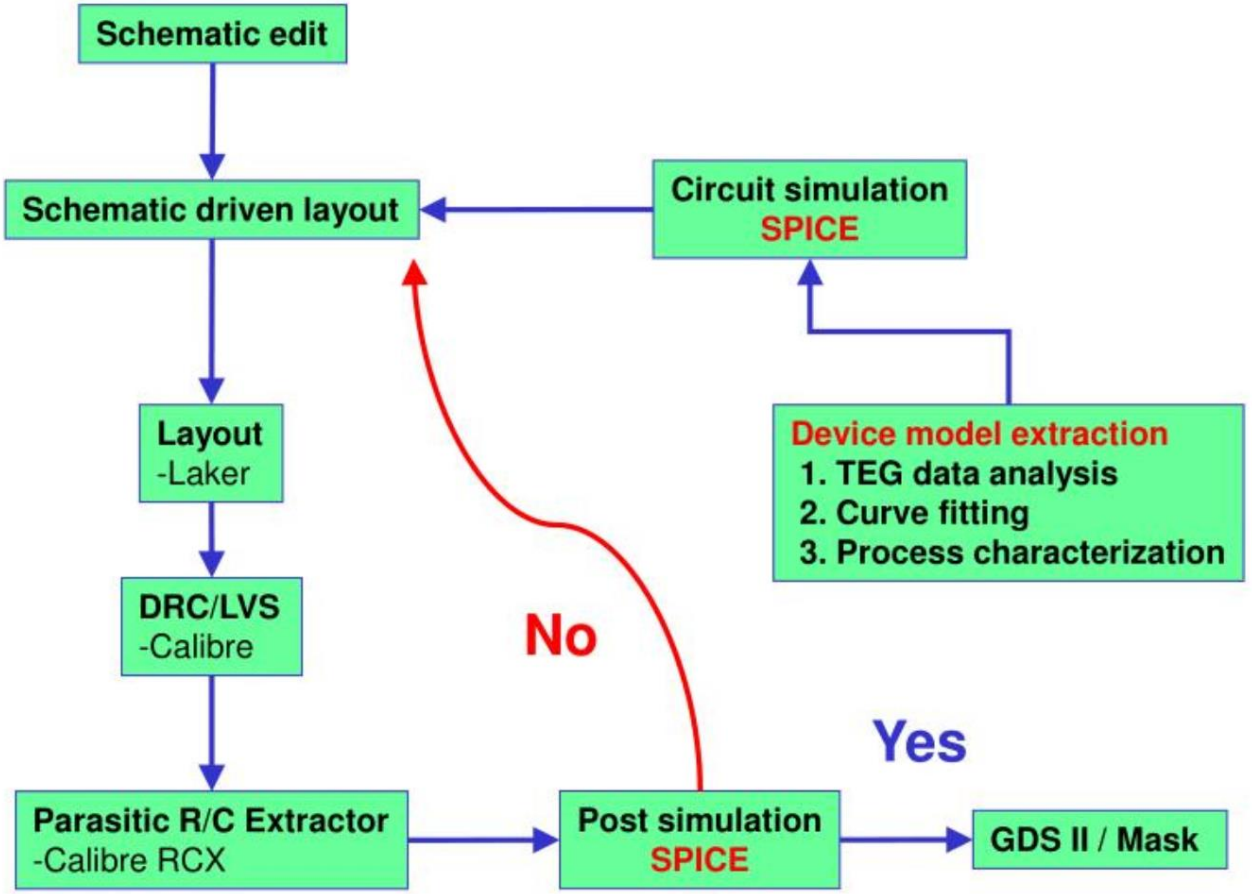


Fig. 1.3-1

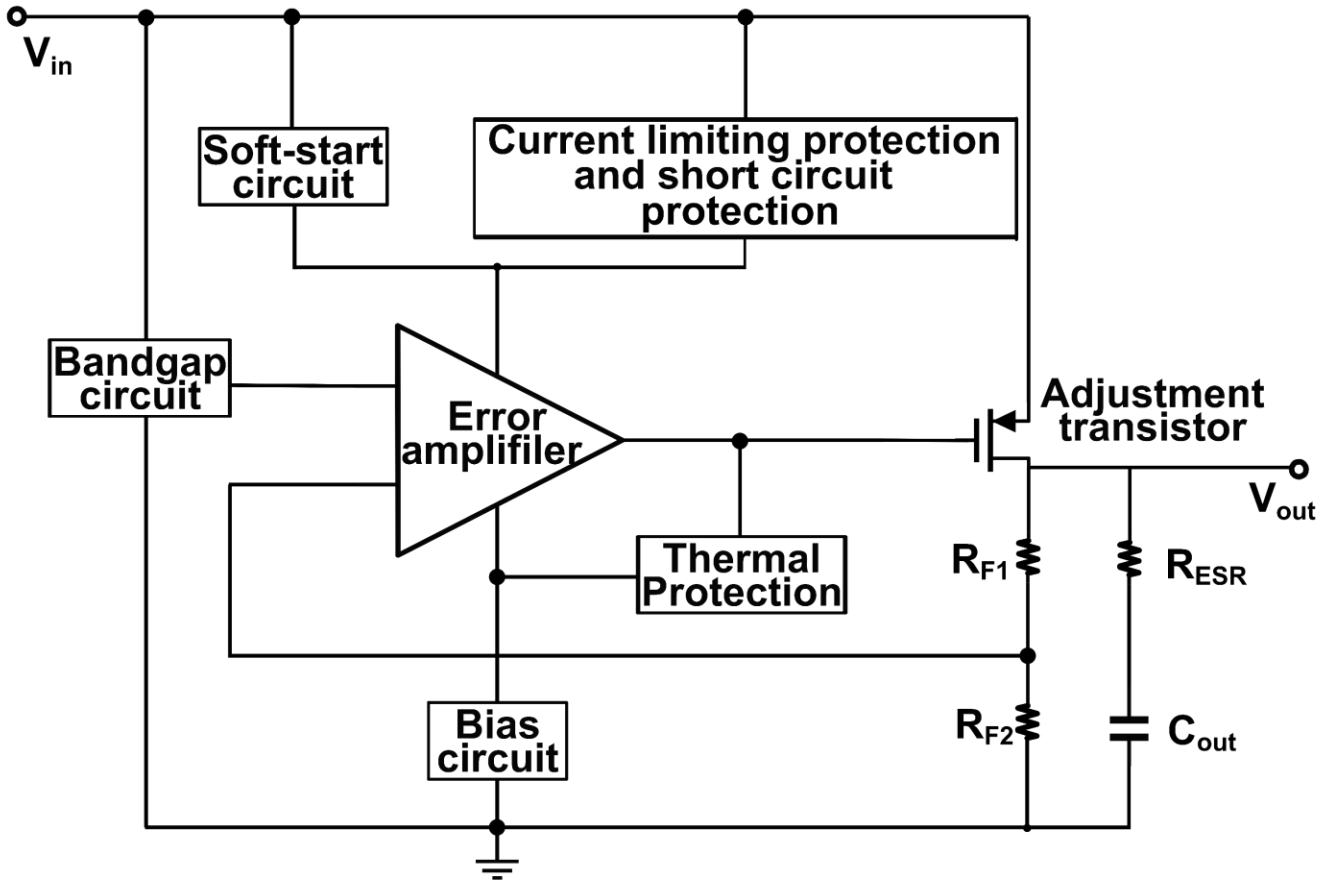
Levels of Abstraction



Analog Design Flow



Analog IC Example

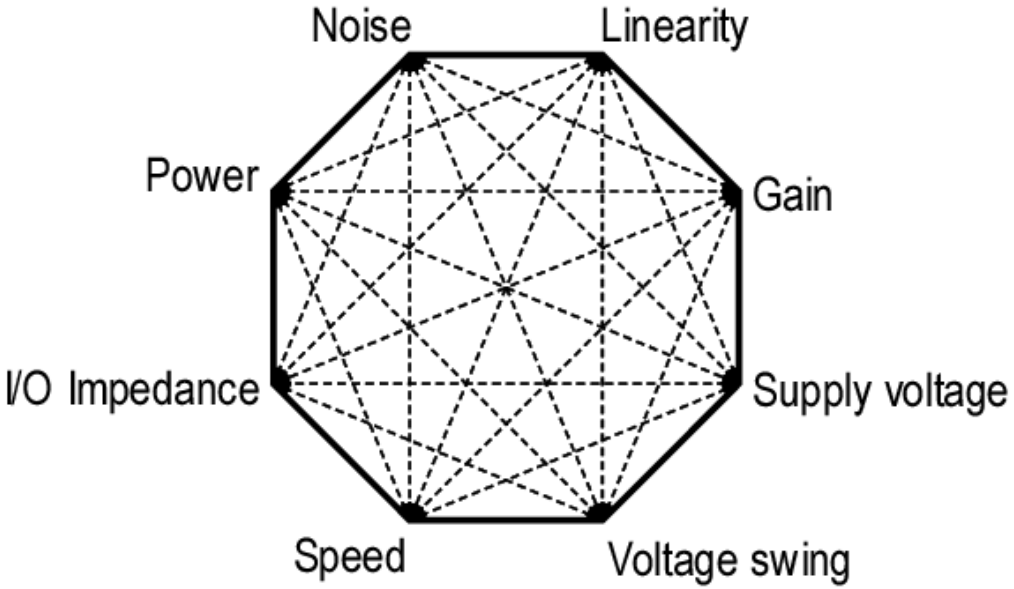


Characteristics of Analog IC

- Analog IC chips usually contain operational amplifiers、voltage and current circuits, etc.
- Performance of **OPAMP** usually determines the whole performance of IC. Therefore, op-amps are not only **basic** circuits but also **core** circuits.
- ASIC usually are contained analog general IC modules

Challenges in Analog Design

- Multi-dimensional **TRADE-OFF**
  - Speed, power dissipation, gain, precision, supply voltage, etc.;



Challenges in Analog Design

- **Sensitive to noise, crosstalk and other interferes;**
- **Heavy second-order effects influence;**
- The design can **rarely be automated**;
  - **Hand-crafted**
- To draw upon experience and intuition;
- Compatible with digital IC technology;

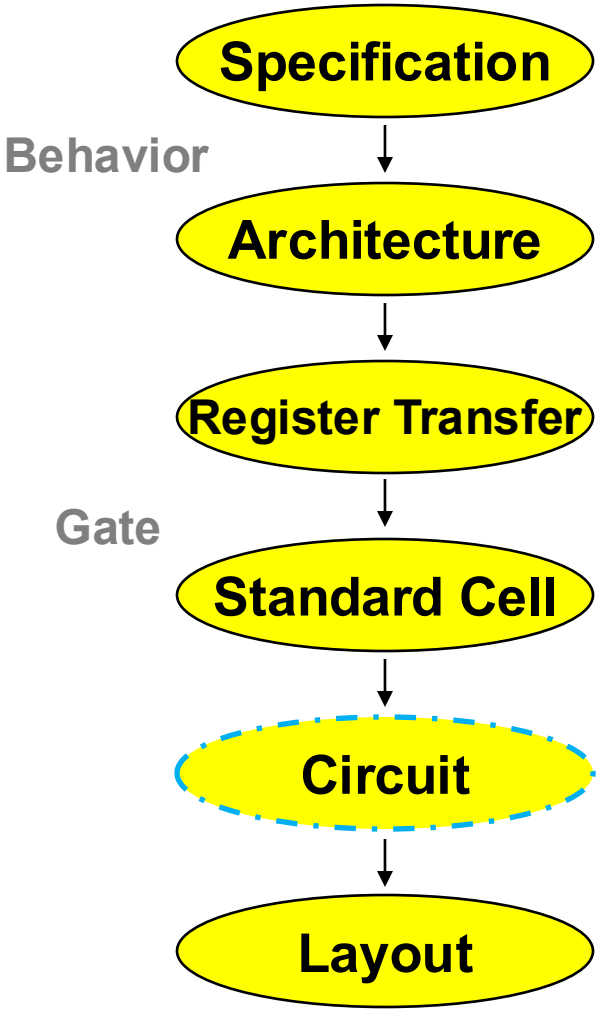
Experience Matters!

Outline

- What is IC and Where it come from:
  - History Class
- Ongoing developments on IC Technology
- Circuits we CARE:
  - Analog Circuits
  - Digital Circuits
- New Era for IC researcher

Digital IC Design Method

- VLSI Design Flow = Creativity + (Design) + Verification
- Design for "X"
  - Design for Test
  - Design for Manufacture
  - Design for yield
- RTL Front-End & GDSII Back-End



Behavior Description (1)

- Algorithm-level behavior description

  adder

  C<=a + b

- 4-bit adder of the Boolean equation

$$s_i = (a_i \oplus b_i) \oplus c_i$$

$$c_{i+1} = a_i b_i + c_i (a_i + b_i)$$

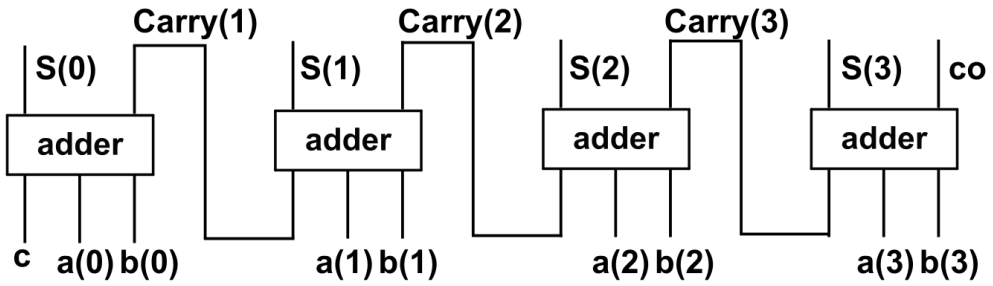
$$i = 0,1,2,3$$

Behavior Description (2)

- Description of a RTL level adder with VHDL

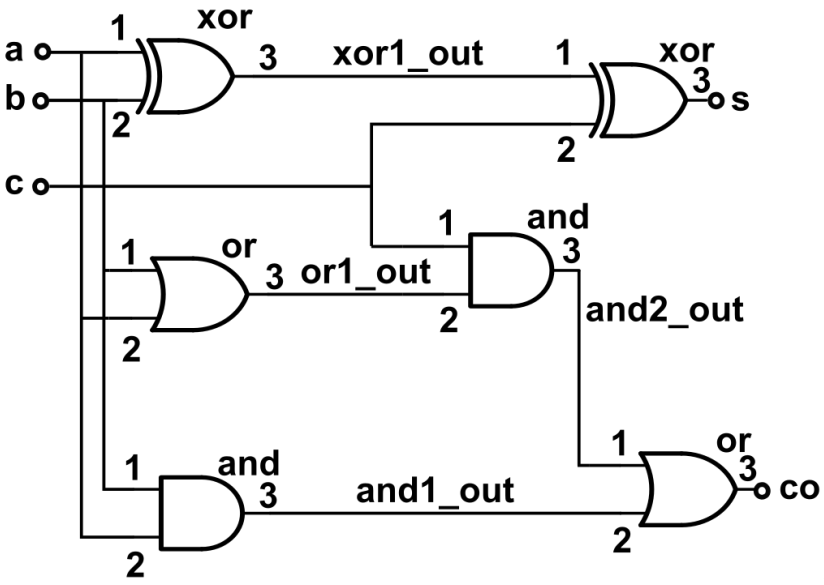
```
entity adder is
port(a, b, c: in std_logic;
       s, co: out std_logic);
end adder;
Architecture rt1 of adder is
begin
        s<=(a xor b) xor c;
        co<=(a and b) or (a and c) or (b and c);
End rt1;
```

Structure description(1)




```
Architecture structure of adder4 is
    component adder
        port(a, b, c: in std_logic;
             s, co: out std_logic);
    end component;
    signal carry: std_logic_vector(1 to 3);
    begin
    a0: adder port map(a(0), b(0), c, s(0), carry(1));
    a1: adder port map(a(1), b(1), carry(1), s(1), carry(2));
    a2: adder port map(a(2), b(2), carry(2), s(2), carry(3));
    a3: adder port map(a(3), b(3), carry(3), s(3), co);
end structure;
```

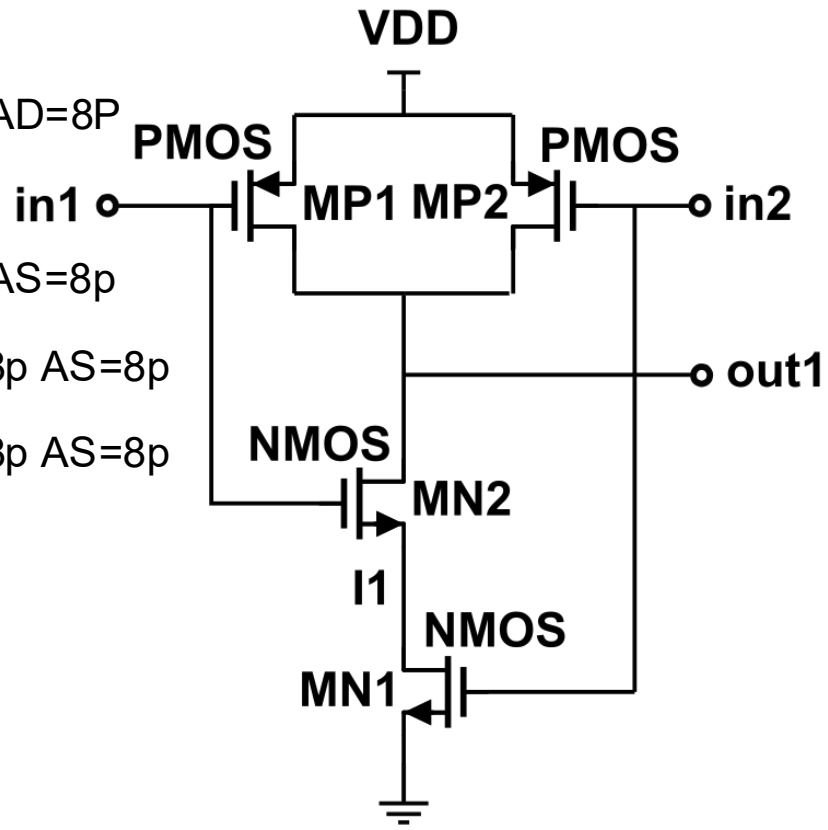
Structure description(2)



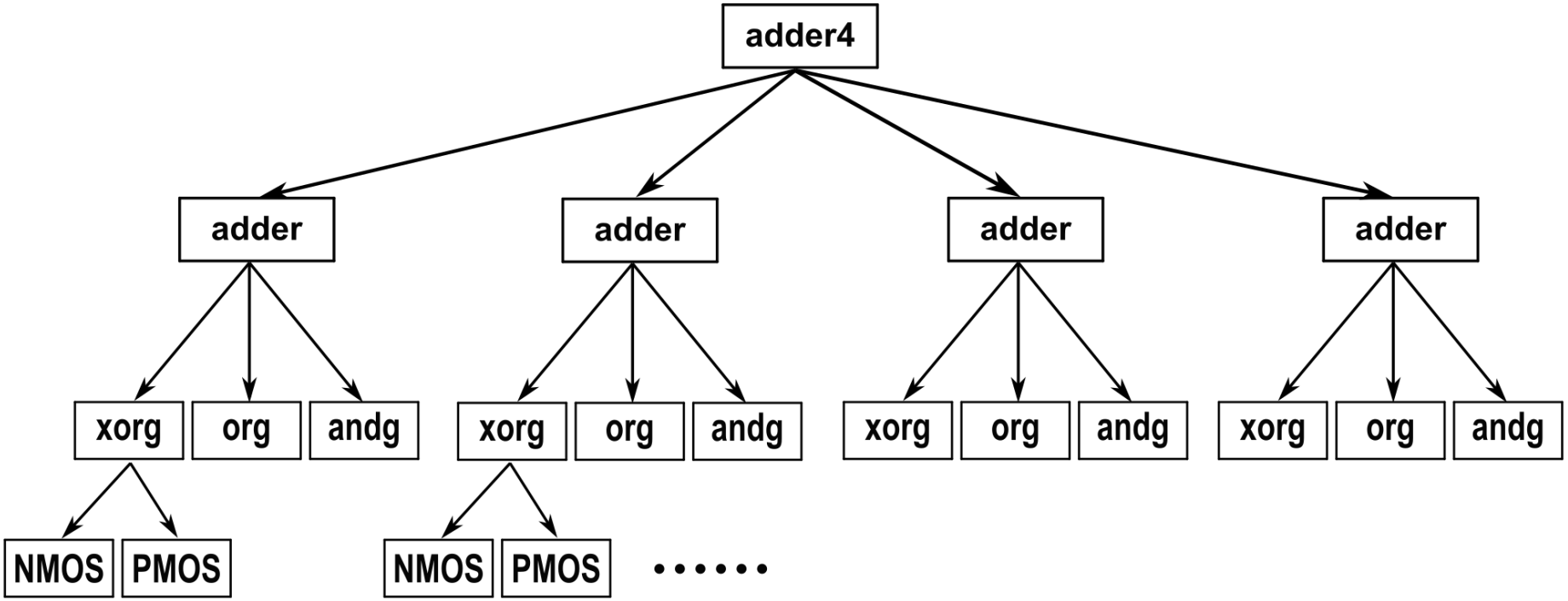

```
Architecture structure of adder is
    component xorg is
        port(in1,in2: in std_logic;
             out1: out std_logic);
    end component;
    component andg is
        port (in1,in2: in std_logic;
              out1: out std_logic);
    end component;
    component org  is
        port(in1, in2: in std_logic;
             out1: out std_logic);
     end component;
signal xor1_out, and1_out,and2_out, or2_out:std_logic;
begin
    xor1: xorg port map (in1=>a, in2=>b, out1=>xor1_out);
    xor2: xorg port map (in1=>xor1_out, in2=>c, out1=>s);
    and1:andg port map(in1=>a, in2=>b, out1=>and1_out);
    or1: org port map(in1=>a, in2=>b, out1=>or1_out);
    and2:andg port map(in1=>c, in2=>or1_out, out1=>and2_out);
     or2: org port map(in1=>and2_out, in2=>and1_out, out1=>co);
end structure;
```

Structure description：netlist

```
SUBCKT nand2 VDD VSS in1 in2 out1
MN1 I1 in2 VSS VSS NMOS NMOS W=8u L=1u AD=8P
      AS=8P
MN2 out1 in1 I1 VSS NMOS W=8u L=1u AD=8p AS=8p
MP1 out1 in1 VDD VDD PMOS W=8u L=1u AD=8p AS=8p
MP2 out1 in2 VDD VDD PMOS W=8u L=1u AD=8p AS=8p
C1 out1 VSS 100PF
C2 I1 VSS 20PF
Cin1 in1 VSS 50PF
Cin2 in2 VSS 50PF
.ENDS
```



Levels of Abstraction



Physical Description

- Describe how to generate the needed structure with using detailed components
- Use **gate-level netlist** to accomplish IC physical design in accordance with IC design ruler,
- Complete different level layouts which are needed in IC fabrication.

Challenges in Digital Design



- Multiple levels of abstractions: Layout to Specification.
- Multiple and conflicting constraints:
  - Low cost and high performance…..
- **Short design time**: Time to Market

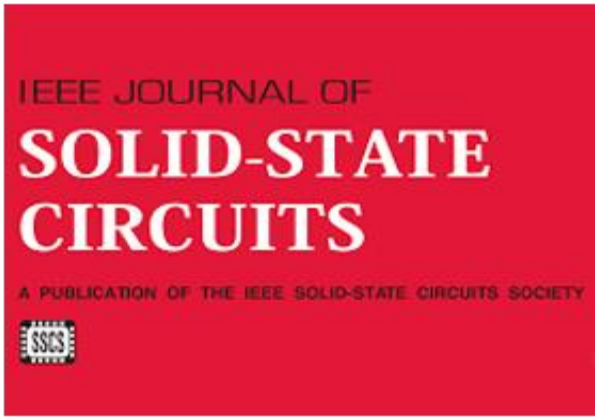
Outline

- What is IC and Where it come from:
  - History Class
- Ongoing developments on IC Technology
- Circuits we CARE:
  - Analog Circuits
  - Digital Circuits
- New Era for IC researcher

ISSSC and JSSC

70+ Years





60+ Years

IEEE International Solid-State Circuits Conference

：是世界学术界和企业界公认的集成电路设计领域最高级别会议，被认为是集成电路设计领域的“**世界奥林匹克大会**”！

ISSSC

- 始于1953年的ISSCC（International Solid-State Circuits Conference）国际固态电路年度会议，通常是各个时期国际上最尖端固态电路技术最先发表之地。由于ISSCC在国际学术、产业界受到极大关注，因此被称为**集成电路行业的奥林匹克大会**。每年吸引了超过3000名来自世界各地工业界和学术界的参加者



百度百科

ISSSC

- 众多集成电路历史上里程碑式的发明都是在这上面上首次披露。比如：
  - 世界上第一个TTL电路 (1962年),
  - 世界上第一个集成模拟放大器电路(1968年),
  - 世界上第一个1kb的DRAM (1970年),
  - 世界上第一个CMOS electronic wristwatch (1971年),
  - 世界上第一个8-bit microprocessor (1974年),
  - 世界上第一个32-bit microprocessor (1981年),
  - 世界上第一个1Mb的DRAM (1984年),
  - 世界上第一个1Gb的DRAM (1995年),
  - 世界上第一个集成 GSM transceiver (1995年),
  - 世界上第一个GHz的微处理器 (2002),
  - 世界上第一个多核处理器 (2005年)等等。

百度百科

Different Areas in ISSCC

ANALOG: Circuits with analog-dominated innovation; amplifiers, comparators, oscillators, filters, references; nonlinear analog circuits; digitally-assisted analog circuits; Sensor interface circuits; analog circuits in sub-10nm scaled technologies

DATA CONVERTERS: Nyquist-rate and oversampling A/D and D/A converters; embedded and application-specific A/D and D/A converters; analog to information conversion; time-to-digital converters.

DIGITAL CIRCUITS and ARCHITECTURES & SYSTEMS*: Digital circuits, building blocks, and complete systems (monolithic, 2.5D, and 3D) for microprocessors, micro-controllers, application processors, graphics processors; digital systems for communications, video and multimedia, cryptography, smart cards, security and trusted computing, accelerators, reconfigurable systems, near- and sub-threshold systems, emerging applications. Digital circuits for intra-chip communication, clock distribution, soft-error and variation-tolerant design, power management (i.e. voltage regulators, adaptive digital circuits, digital sensors), PLLs for digital clocking applications, circuit techniques for neuro-computing, and security circuits (i.e. PUFs, TRNGs, side-channel attack countermeasures, and attack-detection sensors).

IMAGERS, MEMS, MEDICAL, & DISPLAYS: Image sensors and SoCs; automotive, LIDAR, and ultrasonic sensors; MEMS sensor systems; wearable, implantable, ingestible electronics, biomedical SoCs, neural interfaces and closed-loop systems; biosensors, microarrays, and lab-on-a-chip; display electronics, displays with sensing functionality; sensing for AR/VR.

MACHINE LEARNING and AI: Chips demonstrating system, architecture and circuit innovations for machine learning and artificial intelligence: processor architectures, accelerators and digital circuits; mixed-signal, analog, near-sensor and in-sensor processing schemes; all architectures and circuits leveraging near-memory and in-memory computation for AI using volatile or non-volatile memories. Hardware optimizations for new ML models including transformers, hyper-dimensional computing, spiking neural networks, etc.

MEMORY: Static, dynamic, and non-volatile memories for stand-alone and embedded applications; memory/SSD controllers; high-bandwidth I/O interfaces; memories based on phase-change, magnetic, spin-transfer-torque, ferroelectric, and resistive materials; array architectures and circuits to improve low-voltage operation, power reduction, reliability, and fault tolerance; application-specific circuit enhancements within the memory subsystem, including in-memory logic functions and compute.

POWER MANAGEMENT: Power management and control circuits, regulators; switched-mode power converter ICs using inductive, capacitive, and hybrid techniques; energy harvesting circuits and systems; wide-bandgap topologies and gate-drivers; power and signal isolators; robust power management circuits for automotive and other harsh environments; circuits for lighting, wireless power and envelope modulators.

RF CIRCUITS and WIRELESS SYSTEMS:** Building blocks and complete solutions at RF, mm-Wave and THz frequencies for receivers, transmitters, frequency synthesizers, transceivers, SoCs, and SiPs. Innovative circuit-level and system-architecture solutions for established wireless standards and future systems or applications such as radar, sensing, and imaging.

TECHNOLOGY DIRECTIONS: Emerging IC and system solutions for: biomedical, sensor interfaces, analog signal processing, power management, computation (including non-CMOS machine learning), data storage, and communication; non-silicon-, carbon-, organic-, metal-oxide-, compound-semiconductor- and new-device-based circuits; nano, flexible, large-area, stretchable, printable, spintronics, quantum, optical, integrated photonics, and 3D-integrated electronics.

WIRELINE: Receivers/transmitters/transceivers for wireline systems, including backplane transceivers, copper-cable links, chip-to-chip communications, 2.5/3D interconnect, on-chip links, optical links, and silicon photonics; exploratory I/O circuits for advancing data rates, bandwidth density, power efficiency, equalization, robustness, adaptation capability, and design methodology; building blocks for wireline transceivers (such as AGCs, analog and ADC/DAC-based front ends, equalizers, clock generation and distribution circuits including PLLs, line drivers, and hybrids).

Sessions @ ISSCC 2023

SESSION 2 — Monday February 20th, 1:30 PM

Digital Processors

Session Chair: ***Shidhartha Das,*** AMD, Cambridge, United Kingdom
Session Co-Chair: ***Ji-Hoon Kim,*** *Ewha Womans University, Seoul, Korea*

1:30 PM

2.1 DS1 "Zen 4": The AMD 5nm 5.7GHz x86-64 Microprocessor Core
B. Munger[1], K. Wilcox[1], J. Sniderman[1], C. Tung[1], B. Johnson[2], R. Schreiber[3], C. Henrion[2], K. Gillespie[1], T. Burd[4], H. Fair[1], D. Johnson[2], J. White[1], S. McLelland[1], S. Bakke[1], J. Olson[1], R. McCracken[1], M. Pickett[2], A. Horiuchi[2], H. Nguyen[1], T. H. Jackson[2]
[1]AMD, Boxborough, MA; [2]AMD, Fort Collins, CO; [3]AMD, Austin, TX; [4]AMD, Santa Clara, CA

2:00 PM

2.2 A 5G Mobile Gaming-Centric SoC with High-Performance Thermal Management in 4nm FinFET
B-J. Huang, A. Tsai, L. Hsieh, K. Chang, C-J. Tsai, J-M. Chen, E-W. Fang, S-Y. Hsueh, J. Ciao, B. Chen, C. Chang, P. Kao, E. Wang, H. H. Chen, H. Mair, S-A. Hwang
MediaTek, Hsinchu, Taiwan

2:30 PM

2.3 DS1 Amorphica: 4-Replica 512 Fully Connected Spin 336MHz Metamorphic Annealer with Programmable Optimization Strategy and Compressed-Spin-Transfer Multi-Chip Extension
K. Kawamura[*1], J. Yu*[*1], D. Okonogi[1], S. Jimbo[1], G. Inoue[1], A. Hyodo[1], Á. L. García-Arias[1], K. Ando[2], B. H. Fukushima-Kimura[2], R. Yasudo[3], T. V. Chu[1], M. Motomura[1]*
[1]Tokyo Institute of Technology, Yokohama, Japan
[2]Hokkaido University, Sapporo, Japan
[3]Kyoto University, Kyoto, Japan
*Equally Credited Authors (ECAs)

Break 3:00 PM

3:15 PM

2.4 A Fully Integrated End-to-End Genome Analysis Accelerator for Next-Generation Sequencing
Y-L. Chen[*1], C-H. Yang*[*1], Y-C. Wu*[*1], C-H. Lee[2], W-C. Chen[3], L-Y. Lin[3], N-S. Chang[3], C-P. Lin[3], C-S. Chen[3], J-H. Hung[2,4], C-H. Yang[1,2]*
[1]National Taiwan University, Taipei, Taiwan; [2]GeneASIC Technologies, Hsinchu, Taiwan
[3]Taiwan Semiconductor Research Institute, Hsinchu, Taiwan
[4]National Yang Ming Chiao Tung University, Hsinchu, Taiwan
*Equally Credited Authors (ECAs)

3:45 PM

2.5 A 28nm 142mW Motion-Control SoC for Autonomous Mobile Robots
I-T. Lin[1], Z-S. Fu[1], W-C. Chen[2], L-Y. Lin[2], N-S. Chang[2], C-P. Lin[2], C-S. Chen[2], C-H. Yang[1]
[1]National Taiwan University, Taipei, Taiwan
[2]Taiwan Semiconductor Research Institute, Hsinchu, Taiwan

4:15 PM

2.6 VISTA: A 704mW 4K-UHD CNN Processor for Video and Image Spatial/Temporal Interpolation Acceleration
K-P. Lin, J-H. Liu, J-Y. Wu, H-C. Liao, C-T. Huang
National Tsing Hua University, Hsinchu, Taiwan

4:45 PM

2.7 DS1 MetaVRain: A 133mW Real-Time Hyper-Realistic 3D-NeRF Processor with 1D-2D Hybrid-Neural Engines for Metaverse on Mobile Devices
D. Han, J. Ryu, S. Kim, S. Kim, H-J. Yoo
Korea Advanced Institute of Science and Technology, Daejeon, Korea

Conclusion 5:15 PM

SESSION 10 — Tuesday February 21st, 8:30 AM

Pipelined and Noise-Shaping ADCs

Session Chair: ***Nima Maghari,*** *University of Florida, Gainesville, FL*
Session Co-Chair: ***Ping Gui,*** *Southern Methodist University, Dallas, TX*

8:30 AM

10.1 A 1.8GHz 12b Pre-Sampling Pipelined ADC with Reference Buffer and OP Power Relaxations
S-E. Hsieh, T-C. Wu, C-C. Hou
MediaTek, Hsinchu, Taiwan

9:00 AM

10.2 A Single-Channel 2.6GS/s 10b Dynamic Pipelined ADC with Time-Assisted Residue Generation Scheme Achieving Intrinsic PVT Robustness
J. Hao[1,2], M. Zhang[1], Y. Zhang[1,2], S. Liu[2], Z. Zhu[2], Y. Zhu[1], C-H. Chan[1], R. P. Martins[1,3]
[1]University of Macau, Macau, China
[2]Xidian University, Xi'an, China
[3]Instituto Superior Tecnico/University of Lisboa, Lisbon, Portugal

9:30 AM

10.3 A Single-Channel 12b 2GS/s PVT-Robust Pipelined ADC with Critically Damped Ring Amplifier and Time-Domain Quantizer
Y. Cao[1], M. Zhang[1], Y. Zhu[1], C-H. Chan[1], R. P. Martins[1,2]
[1]University of Macau, Macau, China
[2]Instituto Superior Tecnico/University of Lisboa, Lisbon, Portugal

Break 10:00 AM

10:15 AM

10.4 A Rail-to-Rail 12MS/s 91.3dB SNDR 94.1dB DR Two-Step SAR ADC with Integrated Input Buffer Using Predictive Level-Shifting
M. Li[1], C. Y. Lee[1], A. ElShater[1], Y. Miyahara[2], K. Sobue[2], K. Tomioka[2], U-K. Moon[1]
[1]Oregon State University, Corvallis, OR
[2]Asahi Kasei Microdevices, Atsugi, Japan

10:45 AM

10.5 A 25MHz-BW 77.2dB-SNDR 2nd-Order Gain-Error-Shaping and NS Pipelined SAR ADC Based on a Quantization-Prediction-Unrolled Scheme
H. Zhang[1], Y. Zhu[1], C-H. Chan[1], R. P. Martins[1,2]
[1]University of Macau, Macau, China
[2]Instituto Superior Tecnico/University of Lisboa, Lisbon, Portugal

11:15 AM

10.6 A 150kHz-BW 15-ENOB Incremental Zoom ADC with Skipped Sampling and Single Buffer Embedded Noise-Shaping SAR Quantizer
Z. Wang[1], L. Jie[2], Z. Kong[1], M. Zhan[2], Y. Zhong[2], Y. Wang[1], X. Tang[1]
[1]Peking University, Beijing, China
[2]Tsinghua University, Beijing, China

11:45 AM

10.7 A Single-Channel 70dB-SNDR 100MHz-BW 4th-Order Noise-Shaping Pipeline SAR ADC with Residue Amplifier Error Shaping
Y. Zhang[*1,2], J. Hao[1,2], S. Liu[2], Z. Zhu[2], Y. Zhu[1], C-H. Chan*[*1], R. P. Martins[1,3]*
[1]University of Macau, Macau, China
[2]Xidian University, Xi'an, China
[3]Instituto Superior Tecnico/University of Lisboa, Lisbon, Portugal
*Equally Credited Authors (ECAs)

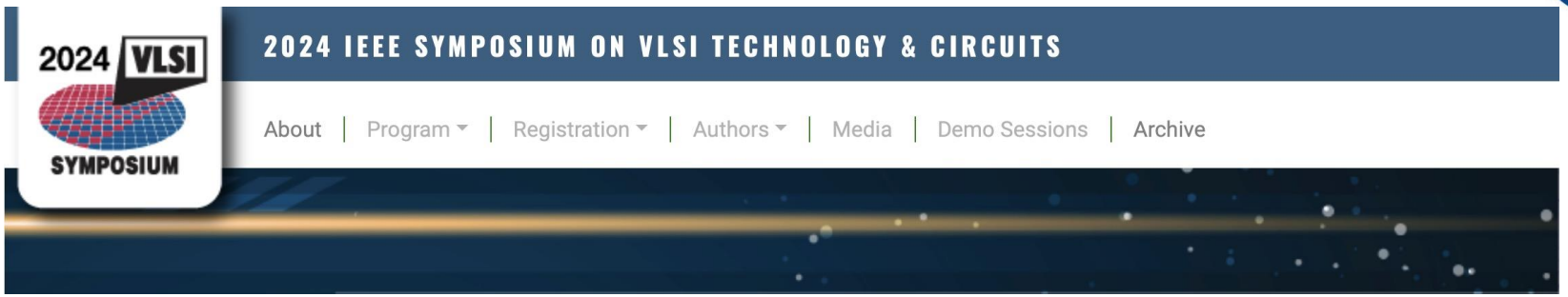
Conclusion 12:15 PM

ISSCC@2023

ISSCC 2023 企业/高校/科研机构

- **企业: 9 家公司 > 2 篇论文**
  - Samsung Electronics, MediaTek, Intel, AMD, Texas Instruments, Sony, TSMC, Renesas Electronics, SK Hynix
- **研究院: 3 间机构 > 2 篇论文**
  - Imec, **Institute of Semiconductors CAS**, **Institute of Microelectronics CAS**
- **高校: 31 所学校 > 2 篇论文**
  - **U. Macau(澳门大学)**, **Tsinghua U. (清华大学)**, KAIST, Nat. Yang Ming Chiao Tung U., **Peking U. (北京大学)**, **UESTC(电子科技大学)**, Tokyo Inst. of Tech., Nat. Tsing Hua U., Nat. Taiwan U., **Southeast U. (东南大学)**, **Zhejiang U. (浙江大学)**, **Fudan U. (复旦大学)**, Ulsan Nat. Inst. of Sci. and Tech., Nat. U. of Singapore, Korea U., Osaka U., Nat. Cheng Kung U., Pohang U. of Sci. and Tech., Seoul Nat. U., Daegu Gyeongbuk Inst. of Sci. and Tech. **(FE: 20)**
  - U. of California San Diego, U. of Illinois at Urbana-Champagne, U. of Toronto, Oregon State U., California Inst. of Tech., Massachusetts Inst. of Tech., U. of Texas at Dallas. Georgia Inst. of Tech., U. of Michigan **(NA: 9)**
  - Delft U. of Tech, Politecnico di Milano **(EU: 2)**

Other Flagship Conferences

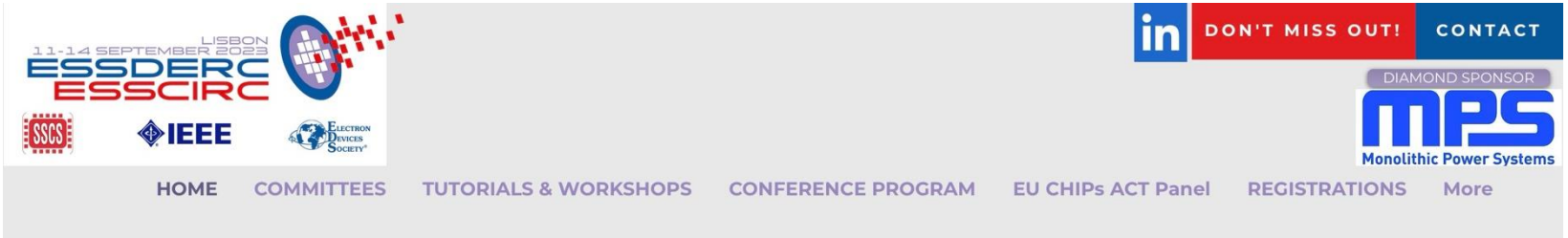


始于1981



始于1996





始于1973



始于2005

集成电路原理与设计

1.导论

谭志超
zhichaotan@zju.edu.cn